

Phase Structure of Field Theories

Ravisankar Rajagopal

PROLOGUE

My very strong desire to work in quantum field did not seem likely to lead to quick publications; but I had already found out that I seemed to be able to get jobs even if I didn't publish anything so I did not worry about 'publish or perish' questions.

— Nobel Lecture, Kenneth Wilson¹

Field theory seems to be nature's choice of language. Be it elementary particles and forces, or electrons swirling through a metal, field theory provides the best description of the system. The reason for this astonishing efficiency of field theories to describe a wide verity of phenomena was understood by Ken Wilson. He was trying to answer the question: *What is a Field Theory?*

By the 1950's, people had understood quantum electrodynamics, through perturbation theory. The theory was organized diagram by diagram, and using clever techniques to render finite answers. Everyone thought that these techniques that swept the infinities under the rug are some gimmick, not physical at all. Wilson was trying to understand if these tricks had any meaning after all. After grappling with lots of simple toy models, he found a simple answer : One has to organize field theory scale-by-scale instead of diagram-by-diagram. This powerful organization principle – goes by the name *Renormalization Group* – lets us define field theory beyond perturbation theory, and we could understand the “dopey procedure” of sweeping infinities under the rug. It goes beyond, combined with Landau's idea of symmetry breaking this tells us about the phase structure of the field theory.

In these notes, we will try to elucidate upon this idea, and try to understand what a field theory really is. We will start with the idea of Renormalization group as Wilson understood it, and then use it to explore a wide array of examples. It turns out that the same organizing principle will tell us how water boils, how magnets behave, how quarks are confined, how general relativity comes from string theory, and what M-theory means. We will explore most of these ideas in some detail in these notes. We will arrange things in the following way: first, we will start with the simplest type of field theories with a global symmetry of $\mathbb{Z}_2$. These will explain anisotropic magnets as well as the boiling of water. Then, we will move on to models with global $O(N)$ symmetry. These will describe isotropic magnets, and opens up the arena for topological phase transitions. Then, we will look at the phases of theories with local symmetries, and see simple models that show confinement. We will study a large array of techniques – perturbative RG, background field method, large N etc. to give a few examples – along the way.

¹The full talk can be found in [1].

According to Polchinski², string theory might be in a similar situation. Although people have obtained reasonable control over string perturbation theory, a full non-perturbative definition is still lacking. The closest we have come maybe is the AdS/CFT correspondence, but that too is not fully understood. Alas, we are too worried about ‘publish or perish’ to think about these issues...

Most of the explanations in the notes are borrowed from a multitude of sources. In order to avoid clutter, I will cite all the sources (along with the relevant sections) at the end of each big section. As suggested by Gary and Dima, I will provide example problems (with solutions) at the end as well. The references from which the problems are taken (or inspired) will be provided at the start of each exercise.

²His article on “What is String Theory” can be found in [2].

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1. The Renormalization Group

I wanted to learn about elementary particles by studying boiling water.

— Alexander Polyakov

In this chapter, we will lay out the general idea of the Renormalization Group (RG), geared towards applications in field theory. The idea of RG is rather philosophical (in the sense that there is no general mathematical prescription one can use in any given context). But fortunately for field theories, one can quantitatively carry out the RG procedure. The framework (even though sometimes is philosophical) is a powerful one, and it puts almost all of physics in perspective.

1.1. Universality

Liquid-Vapor Transition

Take carbon di-oxide in a closed container, at pressure around $p_c \approx 73$ bar and temperature $T \approx 30^\circ$ C. In this range of parameters, both liquid and gas form of carbon di-oxide exists with liquid below and gas above in the container, with a visible boundary between them.



Figure 1: CO₂

The average density is around $\rho \approx 500 \frac{\text{kg}}{\text{m}^3}$. If we put small spheres of varying densities around this value, the lightest of course will float to the top, the heaviest sink, and the ones with in-between densities float in the middle. Now, we start heating up. At around $T_c \approx 31.1^\circ$ C, the boundary between the liquid and gas disappears, and a single “fluid” with density around $\rho_c \approx 470 \frac{\text{kg}}{\text{m}^3}$ remains. All the spheres with lower densities rise up, and heavier ones sink. Increasing the temperature after this do not change the densities anymore. If we plot the density vs temperature curve near $T_c \approx 31.1^\circ$ C, we get the following,

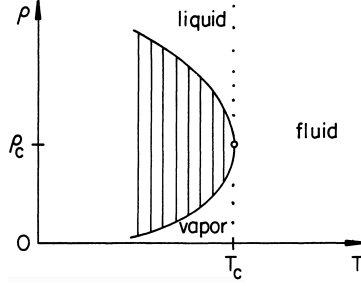


Figure 2: density vs temperature at constant pressure p_c for CO_2

T_c is called the *critical temperature*. If we actually go out and do the experiment, one finds,

$$(\rho - \rho_c) \propto |T - T_c|^\beta \quad \text{with} \quad \beta \approx 0.32 \quad (1.1)$$

One can measure the compressibility, how the density change as we change pressure from the critical value,

$$\kappa = \frac{1}{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T \quad (1.2)$$

and near critical temperature,

$$\kappa \propto |T - T_c|^{-\gamma}, \quad \text{with} \quad \gamma \approx 1.2 \quad (1.3)$$

At the critical temperature, the fluid becomes cloudy as light gets scattered. The reason is that there are density fluctuations at all length scales appearing. That is, there will be air bubbles in the liquid of arbitrary sizes and shapes, and inside these air bubbles, we will have liquid bubbles of arbitrary sizes and shapes, and inside these liquid bubbles we will have air bubbles and so on. No matter how far we zoom out, the system will look the same, as the fluctuations appear at all length scales. This is in contrast with above or below the critical temperature. Below the critical temperature, say the pressure is such that we have a liquid, we will have bubbles inside the liquid, but all of them become invisible if we zoom out far enough. Similarly above the critical temperature density fluctuations gets washed out at long distance scales. As a result, light can get scattered to bubbles that are of the same order in size as the wavelength of light. This only happens at the critical temperature, as that is the only place we have density fluctuations of all scales. This is called critical opalescence³.

There are a number of surprising things in this experiment. First, we usually expect things to be smooth in physics. We would normally expect the density to asymptote to the critical

³Here is a video of this : <https://www.youtube.com/watch?v=DIGdbmJvFUw>. Mixture of methanol and cyclohexane is being heated above to the critical temperature and cooled back down.

value as we get to very large temperatures. Instead, the change found in nature is abrupt, and quantities diverge at the critical point.

The second astonishing fact is the following: no matter what gas we take, at the critical point, it behaves exactly the same as above! This is not just a qualitative statement. For any gas, if one measures the divergences, one gets the same values for β and γ . These numbers seem to be some kind of universal constants, and we call them *critical exponents*.

Paramagnet-Ferromagnet Transition

The surprises do not end here. Take a magnet, and heat it up. As the temperature increases, the molecular dipoles start to jiggle and start to get randomly oriented. The randomness increases as the temperature goes up. What one expects is that the average magnetization decreases slowly – most importantly smoothly – to zero as the temperature goes to infinity.

But once again, the average magnetization becomes zero abruptly at some finite temperature T_c . A rough picture of what happens is as shown below,

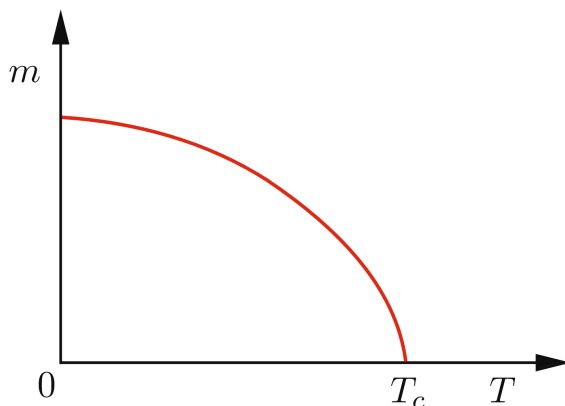


Figure 3: magnetization vs temperature for a magnet

The change in magnetization is not smooth as a function of temperature at T_c . For a three dimensional magnet, we have experimentally found the nature of the curve close to the critical point,

$$m \propto |T - T_c|^\beta \quad \text{with} \quad \beta \approx 0.32 \quad (1.4)$$

If one decides to compute the magnetic susceptibility, that is the change in magnetization as we change the applied magnetic field B ,

$$\chi = N \left(\frac{\partial m}{\partial B} \right)_T \quad (1.5)$$

near the critical temperature, experiments tell us,

$$\chi \propto |T - T_c|^{-\gamma}, \quad \text{with} \quad \gamma \approx 1.2 \quad (1.6)$$

The numbers are exactly the same as that of the liquid vapor system! There are other critical exponents that are measured, for example the heat capacity is found to obey,

$$c \propto |T - T_c|^{-\alpha} \quad \text{with} \quad \alpha \approx 0.11 \quad (1.7)$$

for both the magnet and any gas. At the critical temperature, one can vary the pressure away from the critical value and observe,

$$T = T_c : \quad (\rho - \rho_c) \propto (P - P_c)^{\frac{1}{\delta}} \quad \text{with} \quad \delta \approx 4.8 \quad (1.8)$$

For the magnet, we can do something analogous. Change the external magnetic field from the critical value $B_c = 0$ and we observe,

$$T = T_c : \quad (m - m_c) \propto (B - B_c)^{\frac{1}{\delta}} \quad \text{with} \quad \delta \approx 4.8 \quad (1.9)$$

At the critical point, just like the liquid-vapor system fluctuations of all length scales occur. There will be islands of positive magnetizations of random size and shapes inside which islands of negative magnetizations of random size and shape and so on. Zooming in or out do not statistically change the appearance. Below the critical point, the fluctuations in magnetization will be invisible if we zoom far out, and the same is true above the critical point⁴.

What on Earth does a magnet have to do with boiling water? Seemingly, around a critical point, many vastly different systems behave exactly the same way. At the critical point, any memory of the microscopic details of the system is washed away. This phenomena is known as *universality*.

Understanding what happens when we boil water turns out to be not so simple. There are two deep insights about nature that changed the way we think about phase transitions: Nature is organized by symmetry, and nature is organized by scale.

The fact that nature is organized by symmetry was Landau's insight. He understood that as matter goes from one phase to another, some underlying symmetry of the material change from one class to another. Any change in symmetry class is bound to involve some "violence", as we see in nature.

The next insight, that nature is organized by scale was due to Wilson. He understood that the microscopic details of the theory is not very important when we try to understand

⁴Here is a nice video that visually shows this idea : <https://www.youtube.com/watch?v=MxRddFrEnPc>

physics at scales much larger. As David Tong puts it, “little things affect big things, but not the other way round. Also, little thing very rarely affect very big things. Instead, little things affect slightly bigger thing and so on...”. Wilson re-organized field theory scale-by-scale instead of the perturbative expansion. This idea led to concrete predictions of the critical exponents, that one could check against the experiments.

These two ideas – nature is organized by symmetry and scale – is put together in the frame work called the *Renormalization Group*. This turns out to be how we understand most of modern physics. Far from the pot on the stow, renormalization group is the way we understand anything from the standard model to superconductors to string theory.

The framework of renormalization group is in some sense philosophical. It is not possible to give a general mathematical procedure on how to go about implementing this idea for any problem at hand. Instead, we will work on examples, and understand the general principle through them.

1.2. Defining Phase Transitions

To study collective behavior, the main tool that we have at our disposal is statistical mechanics. In systems with many particles, one assigns a probability for every possible configuration. For fixed temperature systems, the probability of a system being in a given configuration C is,

$$p(C) = \frac{e^{-\beta H(C)}}{Z} \quad (1.10)$$

where Z is the *partition function*, which is the weighted sum over all configurations,

$$Z = \sum_{\{C\}} e^{-\beta H(C)} \quad (1.11)$$

$H(C)$ is the energy of the configuration C , and β is the inverse temperature. We know that almost all the questions about the macroscopic behavior of the system can be expressed in terms of the partition function, and its derivatives. One usually defines the *free energy*,

$$F = -T \ln Z \quad (1.12)$$

Then, all the quantities of interest are expressed as derivatives of F . For example, the heat capacity at constant volume for a gas is,

$$C = -T \frac{\partial^2 F}{\partial T^2} \quad (1.13)$$

The free energy might depend on a multitude of parameters, that we will collectively call *couplings*, and denote by the letter g . These couplings could be a function of temperature,

pressure or external magnetic field. They could also be functions of things that we have no control over, like the microscopic details of the system.

Experiments tell us that at the critical temperature, some of the quantities like the heat capacity or susceptibility are divergent, or in general non-analytic. What this imply is that the free energy itself has non-analyticity at the critical point.

Actually, the above statement is slightly alarming. In the definition of the partition functions, all quantities are analytic. The energy and exponential of the energy are analytic, and so is the sum. Then how can the free energy or the partition function can have any non-analyticity at all? The non-analytic behavior turns out to arise in the *thermodynamic limit*.

From experience, we know that the free energy scales with the volume of space. In general,

$$F = Vf_V + Sf_S + \mathcal{O}(L^{d-1}) \quad (1.14)$$

where V and S are the volume and area of the boundary respectively of the system. We define the quantities,

$$\begin{aligned} f_V &= \lim_{V \rightarrow \infty} \frac{F}{V} \\ f_S &= \lim_{S \rightarrow \infty} \frac{F - Vf_V}{S} \end{aligned} \quad (1.15)$$

to be the bulk and boundary free energies respectively. If the system is defined on a lattice, one would define,

$$f_N = \lim_{N \rightarrow \infty} \frac{F}{N} \quad (1.16)$$

where N is the number of lattice sites. If these limits exists, we then say that a thermodynamic limit exists.

In the thermodynamic limit, the analytic sum in the computation of partition function becomes an infinite sum, and then there is no guarantee that the resulting free energy is analytic. This is what happens at the critical temperature. In real life systems, the number of particles are so large that it almost looks like the thermodynamic limit. This is why we observe apparent non-analyticity in experiments.

The bulk free energy f_V is a function of all the couplings $\mathbf{g}$. As we vary the temperature or external fields, some of the couplings change. The free energy and its derivatives are non-analytic only at the critical point. This tell us that the bulk free energy is analytic almost every where as a function of $\mathbf{g}$. The regions of analyticity in the space of couplings are called *phases*. If there are D couplings, the possible non-analyticities in the $\mathbf{g}$ space are

points, lines, planes etc. Let $D_s < D$ be the dimension of the singular loci. We note that only singular loci of co-dimension (i.e, number of additional dimensions the full space have relative to the singular hyperplane, which is $D - D_s$) one can separate phases, and hence these boundaries are called *phase boundaries*. Goldenfeld gives an example to understand why: to partition a room, one needs to build a wall across it. A pole from the bottom to top won't cut it.

1.3. Effective Field Theories and the Renormalization Group

Universality tells us that at the critical point, the microscopic details of the system are not very important. Therefore, to study phase transitions, we can actually look for simple models, that have the essential properties. In general, this is a difficult task. In the vast majority of systems, this is almost impossible to do⁵. Physicists for the most part will only deal with cases where we can find a simple model. Such models will be called *effective theories*.

How do we find a simple model for the long distance physics in general? There is one simple, powerful trick: guessing! The most difficult part in guessing the effective theory is identifying the relevant degrees of freedom. That is, what quantities characterizes the most important features of the long distance physics? If we can find the degrees of freedom, then we will guess the “Hamiltonian” or the *effective action*. Once we find this, we can find the partition function in the usual way in the canonical ensemble,

$$Z = \sum_C e^{-\frac{S(C)}{T}} \quad (1.17)$$

where S is the effective action, and C an allowed configuration for the degree of freedom.

Now how do we find the effective action? This is also a difficult task. To construct the action, we use symmetry as our guiding principle. We ask: what could be the symmetries of the underlying system? Also, what could be some emergent symmetries at long distances? Then, we write down the most general Hamiltonian/effective action for the degrees of freedom that are allowed by symmetry. Typically, there will be infinitely many terms possible. Very roughly speaking, each of these infinite terms are telling us something about the actual microscopic system. But as we learned from experiments, all these vast majority of microscopic details are not relevant for the physics at very large length scales. How do we identify which terms out of these infinite possible terms are relevant? Here is where the idea of organization by scale comes in. The effective theory that we wrote down will have a range of applicability. Below some certain length scale, this effective model has to become

⁵Here are some examples: The flocking pattern on birds seems like some collective phenomena is taking place, but no one has found a simple model that describes it. Our brain, neural networks, population dynamics etc are examples of things that are collective phenomena that we are unable to explain using simple models.

in-adequate, and the more correct microscopic model has to take over. We take this idea at face value, and flip it over. Say our effective model describes the physics for scales much larger than a . This lowest length scale of the model a will be called an ultra-violet (UV) cut-off. We will be interested in physics at scales $L \gg a$ (called Infra-red (IR)). If $b > a$ is another scale, and $L \gg b$, then the details of what is going on in the range $[a, b]$ should not be at all important for the physics at $L \gg b > a$. Therefore, we will find a way to get rid of the physics that happens in the range $[a, b]$. What happens is that in the process, we will be able to identify from the infinitely many terms possible in the effective theory a handful of terms that will be important at scales $L \gg a$. This is the big idea of the Renormalization Group (RG).

In most of the systems that are of interest to physicists, there is one common feature: things that happen at one place do not affect things that happen really far away. This goes the name of *cluster decomposition principle*, and it just tells us that things interact locally. So, we will spend most of our time discussing local theories. The type of relevant degrees of freedom allowed in local theories are called *fields*.

Lets imagine that we identified the relevant degree of freedom for a system to be a field $\Phi(x)$. Then, locality tells us that the effective action must have the form,

$$S[\Phi(x)] = \int d^d x \mathcal{L}(\Phi(x)) \quad (1.18)$$

We will make one more assumption on the kind of effective actions: this is analyticity in Φ , that is the function $\mathcal{L}$ is assumed to be a Taylor expandable function of Φ . **Coleman-Weinberg is a counter example? I don't understand this yet. Seems like RG generates non-analiticities. This should be a chapter later.**

The field, which hopefully captures the effective degree of freedom should interact. This is key to have emergent phenomena. Cluster decomposition tell us that the most dominant interaction should be nearest neighbor⁶. The nearest neighbor interaction of the field is simple to represent, this is,

$$S_0 = \int d^d x \frac{1}{2} \nabla \Phi(x) \cdot \nabla \Phi(x) \quad (1.19)$$

As said, we will assume this is the most dominant interaction, and controls the strength of all the other possible terms in the effective action at all scales.

We will choose units such that this term has unit magnitude. To find out the units of Φ , take

⁶This is of course another guess, that we will modify depending upon the system. For now, lets go with this to describe the ideas in bit more concrete fashion.

$$\mathbf{x} \rightarrow \mathbf{x}' = \zeta \mathbf{x} \implies \Phi \rightarrow \Phi' = \zeta^{-\Delta_\Phi} \Phi \quad (1.20)$$

Then,

$$\int d^d \mathbf{x} \left(\frac{1}{2} \nabla \Phi \cdot \nabla \Phi \right) \rightarrow \int d^d \mathbf{x}' \zeta^{-d+2+2\Delta_\Phi} \left(\frac{1}{2} \nabla' \Phi' \cdot \nabla' \Phi' \right) \quad (1.21)$$

and choose,

$$-d + 2 + 2\Delta_\Phi = 0 \implies \Delta_\Phi = \frac{2-d}{2} \quad (1.22)$$

Apart from the dominant term, there are infinitely many terms that are possible. It is not possible to describe the form of these in general, unless we know what are the symmetries. We will be as general as possible and simply write,

$$S_{\text{int}} = \int d^d \mathbf{x} \sum_i g_i \mathcal{O}_i[\Phi] \quad (1.23)$$

$\mathcal{O}_i[\Phi]$ is a local quantity constructed out of Φ 's and their derivatives. g_i are called couplings, and they determine the “importance” of that operator.

Along with the effective action comes a cut-off. This is what reminds us where the model breaks down. This is encoded in the action as follows: the effective degree of freedom Φ do not make sense below a length scale a . We can make it mathematical as shown below,

$$\Phi(\mathbf{x}) = \int \frac{d^d \mathbf{k}}{(2\pi)^d} e^{i\mathbf{k} \cdot \mathbf{x}} \Phi_{\mathbf{k}}, \quad \text{with } \Phi_{\mathbf{k}} = 0 \text{ if } |\mathbf{k}| > \Lambda \sim \frac{1}{a} \quad (1.24)$$

Let us now try to explain the procedure of RG in this context. Recall we are interested in physics that happens at much smaller scales in momentum (or much larger scales in length). Let E be the energy scale at which we are probing the system, such that,

$$\frac{E}{\Lambda} \ll 1 \quad (1.25)$$

Then comes Wilson's idea that “small things very rarely affect very big things”. To know the physics at scale E , the details of what happens in the scale $\left[\frac{\Lambda}{\zeta}, \Lambda\right]$ where $\zeta > 1$ is not relevant at all (assuming $E \ll \frac{\Lambda}{\zeta}$). We might as well use a theory with lower cut-off $\frac{\Lambda}{\zeta}$ and get the same physics at scale E .

Should we just re-scale all momenta by $\mathbf{k} \rightarrow \frac{\mathbf{k}}{\zeta}$? This indeed does change the cut-off $\Lambda \rightarrow \frac{\Lambda}{\zeta}$, but this is rather trivial. We just changed units. What we want is something more –

we want to throw away all the information in the energy window $\left[\frac{\Lambda}{\zeta}, \Lambda\right]$. To do this, we split our field as follows,

$$\Phi(\mathbf{k}) = \Phi_+(\mathbf{k}) + \Phi_-(\mathbf{k}) \quad (1.26)$$

where,

$$\Phi_+(\mathbf{k}) = \begin{cases} \Phi(\mathbf{k}) & \text{if } \Lambda > |\mathbf{k}| > \frac{\Lambda}{\zeta} \\ 0 & \text{if } 0 \leq |\mathbf{k}| \leq \frac{\Lambda}{\zeta} \end{cases} \quad \text{and} \quad \Phi_-(\mathbf{k}) = \begin{cases} 0 & \text{if } \Lambda > |\mathbf{k}| > \frac{\Lambda}{\zeta} \\ \Phi(\mathbf{k}) & \text{if } 0 \leq |\mathbf{k}| \leq \frac{\Lambda}{\zeta} \end{cases} \quad (1.27)$$

The Φ_+ part is not of interest to us, so we can throw them away. Therefore, in computing the partition function, we integrate out the Φ_+ modes,

$$Z = \int \mathcal{D}\Phi_- \left[\int \mathcal{D}\Phi_+ e^{-S_\Lambda[\Phi_+, \Phi_-]} \right] \equiv \int \mathcal{D}\Phi_- e^{-S_{\frac{\Lambda}{\zeta}}[\Phi_-]} \quad (1.28)$$

where,

$$e^{-S_{\frac{\Lambda}{\zeta}}[\Phi_-]} := \int \mathcal{D}\Phi_+ e^{-S_\Lambda[\Phi_+, \Phi_-]} \quad (1.29)$$

$S_{\frac{\Lambda}{\zeta}}[\Phi_-]$ is some new effective action that only involve the modes with $|\mathbf{k}| < \frac{\Lambda}{\zeta}$.

What is the action $S_{\frac{\Lambda}{\zeta}}$? Well, the original S_Λ was guessed based on general assumptions, and we included all possible terms that were permissible. In general, S_Λ must also have the same form. The only difference is that the coefficients would be different,

$$S_{\frac{\Lambda}{\zeta}}[\Phi_-] = \int d^d \mathbf{x} \left[\frac{\gamma}{2} \nabla \Phi_- \cdot \nabla \Phi_- + \sum_i g'_i \mathcal{O}_i[\Phi_-] \right] \quad (1.30)$$

Now we can ask: how are g' related to the old g ? Maybe integrating out information that we do not care about will reduce the importance of some of the terms by lowering the coefficients? We should therefore compare the coefficients of S_Λ to that of $S_{\frac{\Lambda}{\zeta}}$. But this is not possible at the moment, because these theories have different units. That is, we have,

$$S_\Lambda = \int^\Lambda \frac{d^d \mathbf{k}}{(2\pi)^d} \left(-\frac{\mathbf{k}^2}{2} \Phi(\mathbf{k}) \Phi(-\mathbf{k}) + \sum_i g_i \mathcal{O}_i[\Phi] \right) \quad (1.31)$$

and,

$$S_{\frac{\Lambda}{\zeta}} = \int^{\frac{\Lambda}{\zeta}} \frac{d^d \mathbf{k}}{(2\pi)^d} \left(-\gamma \frac{\mathbf{k}^2}{2} \Phi_{-}(\mathbf{k}) \Phi_{-}(-\mathbf{k}) + \sum_i g'_i \mathcal{O}_i[\Phi_{-}] \right) \quad (1.32)$$

Changing units is easy. Let us take the theory with cut-off $\frac{\Lambda}{\zeta}$, and simply re-scale,

$$\mathbf{k} \rightarrow \mathbf{k} = \frac{\mathbf{k}'}{\zeta} \quad \text{or} \quad \mathbf{x} \rightarrow \mathbf{x} = \zeta \mathbf{x}' \quad (1.33)$$

This gives,

$$S_{\frac{\Lambda}{\zeta}} \rightarrow \int d^d \mathbf{x}' \zeta^d \left(\zeta^{-2} \frac{\gamma}{2} \nabla' \Phi \cdot \nabla' \Phi + \sum_i g'_i \mathcal{O}_i[\Phi] \right) \quad (1.34)$$

There is a small subtlety here. In the original theory, we said that we choose units such that the most dominant term – the nearest neighbor term – has unit magnitude. Therefore,

$$\mathbf{x} \rightarrow \mathbf{x}' = \frac{\mathbf{x}}{\zeta} \implies \Phi_{-} \rightarrow \Phi'_{-} = \zeta^{\Delta_{\Phi}} \Phi_{-} \quad (1.35)$$

Therefore, under proper unit change, we get,

$$S_{\Lambda} \rightarrow \int d^d \mathbf{x}' \left(\frac{\gamma}{2} \nabla' \Phi'_{-} \cdot \nabla' \Phi'_{-} + \zeta^d \zeta^{-\Delta_i} \sum_i g'_i \mathcal{O}_i[\Phi'_{-}] \right) \quad (1.36)$$

where we took,

$$\mathcal{O}_i[\Phi] \rightarrow \mathcal{O}_i[\Phi'_{-}] = \zeta^{+\Delta_i} \mathcal{O}_i[\Phi_{-}] \quad (1.37)$$

under change of units. We are almost ready to compare the couplings. There is still some difference in units, now due to the integrating out procedure. We must re-scale,

$$\Phi'_{-} \rightarrow \tilde{\Phi} = \sqrt{\gamma(\zeta)} \Phi'_{-} \quad (1.38)$$

This changes,

$$\mathcal{O}_i[\Phi'_{-}] \rightarrow \frac{1}{(\sqrt{\gamma(\zeta)})^{N_i}} \mathcal{O}_i[\tilde{\Phi}] \quad (1.39)$$

where N_i is the number of Φ_{-} 's in the term $\mathcal{O}_i$. We now have two theories in the same units,

$$\int d^d \mathbf{x} \left(\frac{1}{2} \nabla \Phi \cdot \nabla \Phi + \sum_i g_i \mathcal{O}_i[\Phi] \right) \quad (1.40)$$

vs

$$\int d^d \mathbf{x} \left(\frac{1}{2} \nabla \tilde{\Phi} \cdot \nabla \tilde{\Phi} + \sum_i \zeta^{d-\Delta_i} \frac{g'_i}{(\gamma(\zeta))^{\frac{N_i}{2}}} \mathcal{O}_i[\tilde{\Phi}] \right) \quad (1.40)$$

These are two theories that describe physics below scale Λ . The second one contains less information than the first, due to the integrating out procedure. But if the idea that nature is organized by scale is correct, then both theories should be identical to describe physics at scale $E \ll \Lambda$. Therefore, we can write in general a theory,

$$S_\Lambda = \int d^d \mathbf{x} \left(\frac{1}{2} \nabla \Phi \cdot \nabla \Phi + \sum_i g_i(\zeta) \mathcal{O}_i[\Phi] \right) \quad (1.41)$$

where,

$$g_i(\zeta) = \frac{\zeta^{d-\Delta_i}}{[\gamma(\zeta)]^{\frac{N_i}{2}}} g'_i(\zeta), \quad \text{with } g_i(\zeta = 1) = g_i \quad (1.42)$$

describes the physics at scale $E \ll \Lambda$ for any $\zeta > 1$. As we increase ζ , more and more short distance physics is thrown away. This change in the couplings under RG is called the *Renormalization Group Flow*.

Summary of the RG Procedure

What we did above are the steps of the Renormalization group (RG). Let us re iterate what these steps are,

1. Integrate out high momentum modes, i.e, $\frac{\Lambda}{\zeta} < |\mathbf{k}| < \Lambda$, where $\zeta \geq 1$
2. Change units of space in the resulting effective action to $\mathbf{x}' = \frac{\mathbf{x}}{\zeta}$
3. Change the units of the field φ to the same units as the original action. That is, ensure the dominant term has unit normalization.

With these three steps, we can understand how the couplings change when we sort of “zoom out” and look at larger length scales.

Meaning of Universality

Our work is still not done, as we still have infinite number of couplings. Our hope is that when we zoom out, most of these couplings will become irrelevant for the physics at large length scales. This sort of is the lesson from universality, that microscopic details do not matter. We hope that majority of the infinite couplings are those “microscopics” that do not change the macroscopics.

To figure out which of the couplings are important, let us do the following. Imagine that we are integrating out only a small sliver of modes that lie in the infinitesimal region, $[\Lambda, \Lambda + d\Lambda]$. Then, it is reasonable to assume that after the first step of RG (integrating out) the couplings only change by an infinitesimal amount,

$$g'_n = g_n + \delta g_n(\zeta) \quad (1.43)$$

Therefore, we have the running,

$$g_i(\zeta) = \frac{\zeta^{d-\Delta_i}}{[\gamma(\zeta)]^{\frac{N_i}{2}}} (g_i + \delta g_i(\zeta)) \quad (1.44)$$

In general as the couplings flow, two things can happen⁷: the couplings could just flow to larger and larger values, essentially flowing to infinity as ζ increases. Or, it could flow to some finite point $\mathbf{g}_c$ in the space of all couplings (will henceforth be referred to as the theory space) as ζ going to infinity. The second case is immediately more interesting. What it tell us is that if we somehow start at that values of coupling constants, then RG does not change anything. Such points in the space of couplings is called a *fixed point*.

What is special about the fixed point? Recall that the second step of RG involved scaling space. Since a fixed point in theory space do not change under RG, a fixed point therefore cannot have any characteristic length scales! Therefore, of all the couplings that has any dimensions, they only have one option at the fixed point: scale to zero! A field theory without a length scale is called *conformal field theory*.

A natural length scale in any field theory is the *correlation length*. This is the length scale in which the fluctuations in the field are correlated. We started off with nearest neighbor interactions being the most dominant term of the effective theory. So in general, it is reasonable to expect that the field fluctuations far away to each other will be uncorrelated. The distance scale that determines this is the correlation length, ξ . At a fixed point, we cannot have any length scales. What that means is that at the fixed point,

$$\xi = 0 \quad \text{or} \quad \infty \quad (1.45)$$

Zero correlation length imply that there are no fluctuations from the average value of the field configuration. This is rather boring. The infinite correlation length is much more interesting. We would have fluctuations happening at all length scales, which would have a power law correlation. This is exactly one observes at the critical point – fluctuations at all length scales. So, a fixed point with $\xi = \infty$ will be identified with a critical point.

⁷there is a third thing, that is the values just go in a loop as ζ increases. These are called limit cycles, and we will ignore them. Most theories of interest apparently do not have limit cycles.

Let us imagine sitting infinitesimally close to a fixed point, and ask : what happens to the couplings under RG? Let us start easy, and ignore the first step of RG, where we integrate out the high momentum modes, and only look at the change due to the re-scaling of position. We have,

$$g_n(\zeta) = \zeta^{d-\Delta_n}(g_{n,c} + \varepsilon_n) \quad (1.46)$$

If $d < \Delta_n$ then the coupling will simply scale to zero as we increase ζ . Such terms are therefore said to be *irrelevant* for the long distance physics. When $d > \Delta_n$, the coupling increase as we coarse grain, and therefore that coupling is said to be *relevant* for the long distance physics. When $d = \Delta_n$, nothing can be said, the coupling is naively important at all scales. We should look at the changes due to the first step in RG to say anything more. Such couplings will be called *marginal*. These are the only couplings at a critical point that can have a finite, non-zero value since they do not scale. If δg – the change due to step 1 of RG – of a marginal operator is negative/positive slightly away from the fixed point, then as we call the coupling *marginally irrelevant/relevant*. If δg turns out to be zero, then the coupling is truly marginal.

The lesson from RG is then the following: the long distance physics depends on the microscopic details only through a few relevant and marginal couplings. One can see the effect of leading irrelevant couplings if we measure phenomena at small enough scales. At the critical point, all the relevant couplings has to go to zero. This explains universality. If two distinct microscopic theories have same symmetries, then they share the same effective field theory but with different couplings. But at the critical point, almost all of the couplings become zero, implying the same behavior for both the theories at the critical point.

Quantum Field Theory

The same logic carries out to quantum field theory. A QFT in d spacetime dimensions is a classical statistical field theory in d space dimensions after a Wick rotation. One usually think about QFT as a fundamental theory, that is supposed to hold for arbitrary small distance scales. A more humble way is actually to acknowledge the cut-off. This way, one gets to understand the renormalization of QFT in a more physical way, rather than simply a trick.

A QFT is said to be renormalizable if the interactions are all relevant or marginal. This way, one gets to interpolate the the theory to arbitrary large energy scales. But this is only rarely true. In many theories (QED for example), the marginal interaction becomes irrelevant due to quantum effects (analogue of thermal effects). What this means is that one can only make sense of QED to arbitrary high energies if there are no marginal interactions as well. But this leads to a boring, non interacting theory. In the eyes of

Wilsonian RG, this is no big deal. QED should be viewed as an effective theory, that cannot be extrapolated to large energy scales.

In contrast, the relevant terms (which are renormalizable) are the real issue in interpreting a QFT. To understand why, let us take a simple scalar field theory with a mass term,

$$S = \int d^d \mathbf{x} ((\partial\varphi)^2 + m^2\varphi^2) \quad (1.47)$$

If we assume that the gradient term is the most dominant term, then the scaling (or dimensions) of φ has to be,

$$\mathbf{x} \rightarrow \zeta \mathbf{x} \implies \varphi \rightarrow \zeta^{\frac{2-d}{2}} \varphi \quad (1.48)$$

This tells us,

$$m^2(\zeta) = \zeta^2 m_0^2 \quad (1.49)$$

the mass term is relevant. Therefore, as we do RG, the mass term becomes larger and larger, and at some point will dominate $(\partial\varphi)^2$ term. What this tells us is the the field φ will become non dynamical at low energies. It would not propagate as it is too heavy, and will not be present in any of the asymptotic states that we could detect in an experiment. Another way to understand this is the spectrum shown below,

$$\text{picture} \quad (1.50)$$

The probe energy is too low to excite a φ particle. What this means is that we cannot have mass terms in QFT! The mass at the cut-off scale has to look like,

$$m_0^2 = \lambda_m \Lambda^2 \quad (1.51)$$

simply because cutoff is the only dimensionful constant that we have. If λ_m is of order 1, then the cut-off mass itself is large. It becomes even larger as we go to IR. If we were to see the φ particles in the IR, then λ_m have to be fine tuned to very small value. λ_m recall is a microscopic parameter. Fine tuning this to a very small value is un-natural.

Most interesting quantum field theories are natural. The masses are prohibited by gauge symmetry or chiral symmetry. There is one exception, which is the Higgs. The Higgs field is a scalar field with a mass term, and therefore is un-natural, and would require fine tuning (meaning we need better explanations). This is still an open problem.

In thermal field theories, we sometimes have external parameters like temperature or magnetic field or pressure that we can change using a knob. Therefore, we can afford to have relevant parameters in the theory as long as there are enough external knobs we can

turn to fine tune these relevant parameters. An example we will see is that of the scalar field, but this time describing the liquid-vapor transition. The mass term turns out to be tunable by changing the temperature, and we can tune it to zero to hit the critical point.

What about other relevant interactions? A similar story is true here as well. These will have couplings that is equal to some number times some power of the cut off scale, which is large. Therefore, the interactions will be too strong, and we will be unable to use φ as the effective degree of freedom. Some new effective degree of freedom should emerge out of this strong coupling, and we should look for a new effective theory, even at the cut-off scale.

There is one more possibility, marginal interactions turning relevant due to quantum/thermal corrections. A marginal interaction is just a dimensionless number, of order 1. This interaction becoming relevant means that the interaction will eventually slowly become large as we flow to IR. Therefore, a theory with marginally relevant interactions are still good from the cut-off scale to some low energy scale below which the interaction becomes strong and we will have to look for a new theory below that scale. An example is QCD, where Yang-Mills theory is good for a range of energy scales, and at very low energies one have to look for a new effective theory. This is better than having relevant couplings, where we have to look for a new effective theory at the cut-off scale itself.

1.4. Exercises

Problem 1 : The purpose of this problem is to see a simple example where taking the infinite volume limit introducing a singular behavior in an otherwise regular system. The problem is inspired from the introduction of Wilson's paper [3].

1. Consider a particle moving according to Aristotle's law,

$$\dot{x}(t) = -\frac{dV(x)}{dx} \quad (1.52)$$

in a double well potential,

$$V(x) = \frac{1}{4}(x^2 - 1)^2 \quad (1.53)$$

Find the particle's position after a time T , if it starts from $x(t = 0) = x_0$.

2. Plot the $x(T; x_0)$ as a function of x_0 for different values of T . Does the plot become more and more steeper at $x_0 = 0$ as we increase T ?
3. From the exact solution, argue that for $T \rightarrow \infty$ (the infinite volume limit) the solution becomes discontinuous.

Solution

1. We have,

$$T + C = -\int \frac{dx}{x(x^2 - 1)} = \frac{1}{2} \ln \left(\frac{x^2}{1 - x^2} \right) \quad (1.54)$$

Using the initial condition, we can solve for $x(T)$,

$$x(T; x_0) = \Theta(x_0) \frac{x_0}{\sqrt{x_0^2 + (1 - x_0^2)e^{-2T}}} - \Theta(-x_0) \frac{x_0}{\sqrt{x_0^2 + (1 - x_0^2)e^{-2T}}} \quad (1.55)$$

The appearance of the theta function is due to the fact that the particle will fall to the right if it starts right to $x = 0$ and to the left otherwise.

- 2.

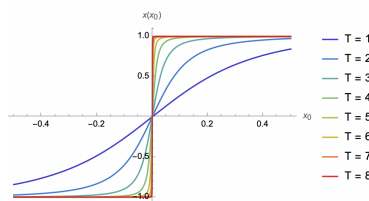


Figure 4: $x(x_0)$ vs x_0 for varying values of T

3. The discontinuity for $T \rightarrow \infty$ is straightforward, stemming from the theta function. We have $x(\infty; x_0 > 0) = 1$ and $x(\infty; x_0 < 0) = -1$.

Problem 2 : This problem is an elaboration of section 2.2 of Goldenfeld[4].

1. Consider a gas of positive charges each with charge q distributed with a uniform density ρ_0 over some spherical volume. Show that one cannot take the thermodynamic limit by calculating the energy of the configuration.
2. Now, let us consider the case where both $\pm q$ charges with equal density is present in the system, with equal uniform density ρ_0 . Now place say, a positive test charge Q at the origin. Positive ions will be repelled from the origin and negative ions attracted towards the origin. Let the effect of all this be a potential at $\mathbf{r}$ be $\Phi(r)$. Argue that the new densities of the positive and negative species can be approximated to be,

$$\rho_{\pm}(r) = \rho_0 \exp(\mp \beta q \Phi(r)) \quad (1.56)$$

What is the total charge density now?

3. Approximate the charge density to linear order when $T \gg q\Phi$. Find the potential in this approximation. This is known as the Debye-Hückel model of screening.
4. Argue that if the background is neutral, then a thermodynamic limit is possible.

Solution

1. The potential energy for positive charges of magnitude q is,

$$U(r) = \frac{q^2}{r} \quad (1.57)$$

The energy inside a spherical region with radius R can be found in the following way. Take an infinitesimal blob of charge at some radius r . This blob has a charge $q\rho_0 d^3r$. Gauss' law tells us that this charge only feels the charges inside the sphere of radius r . More over, we can think of all of the charge inside the sphere of radius r to be sitting at the origin. That is, the blob experiences a potential due to a charge equal to $\frac{4}{3}\pi r^3 \rho_0 q$ sitting at the origin. The blob has the energy,

$$\frac{4}{3}\pi r^3 \rho_0 q \times q\rho_0 d^3r \quad (1.58)$$

Then, the total energy inside the sphere is,

$$E(R) = 4\pi \int_0^R dr r^2 \frac{(q\rho_0)(\frac{4}{3}\pi r^3 \rho_0)}{r} = \frac{(4\pi)^2}{15} (q\rho_0)^2 R^5 \quad (1.59)$$

Energy per unit volume is,

$$\mathcal{E}(R) = \frac{E(R)}{\frac{4}{3}\pi R^3} = \frac{4\pi}{5}(q\rho_0)^2 R^2 \quad (1.60)$$

In the thermodynamic limit, $R \rightarrow \infty$, and the energy per unit volume is not finite.

2. The energy cost of any charge at any distance r is,

$$E(r) = \pm q\Phi(r) \quad (1.61)$$

We can then assume the background charges arrange themselves according to Boltzmann distribution,

$$\rho_{\pm}(r) = \rho_0 \exp(\mp \beta q\Phi(r)) \quad (1.62)$$

The total charge density is,

$$q\rho = q\rho_0(\rho_+ - \rho_-) = -2\rho_0 q \sinh(\beta q\Phi(r)) \quad (1.63)$$

3. When $T \gg q\Phi$, we have,

$$\rho(r) \approx -2\rho_0 \beta q\Phi(r) \quad (1.64)$$

We can write the Poisson equation in the presence of the test charge to be,

$$\begin{aligned} \nabla^2 \Phi(r) &= 4\pi(Q\delta^3(\mathbf{r}) + q\rho(r)) \\ &\approx 4\pi(Q\delta^3(\mathbf{r}) - 2q^2\rho_0\beta\Phi(r)) \end{aligned} \quad (1.65)$$

Define,

$$\lambda_D^2 = \frac{1}{8\pi q^2 \rho_0 \beta} \quad (1.66)$$

this is known as the Debye-Hückel screening length, for reasons that will be clear now. We have,

$$\left(\nabla^2 + \frac{1}{\lambda_D^2} \right) \Phi(r) = 4\pi Q\delta^4(\mathbf{r}) \quad (1.67)$$

This is something we know, the solution is the Yukawa potential,

$$\Phi(r) = -Q \frac{\exp\left(-\frac{r}{\lambda_D}\right)}{r} \quad (1.68)$$

4. We can compute the energy as in part 1. Since the potential due to a blob of charge die out quickly outside the screening length, as far as blobs outside the radius λ_D are concerned, the blob at the origin is non-existent. Therefore, the total energy can be thought to be the energy of a sphere of size λ_D times the number of such spheres. This is,

$$E \sim \frac{\text{Vol}}{\frac{4}{3}\pi\lambda_D^3} E(\lambda_D) \propto \frac{\text{Vol}}{\frac{4}{3}\pi\lambda_D^3} (q\rho_0)^2 \lambda_D^5 \quad (1.69)$$

The volume factorizes, and therefore the energy per unit volume is finite as we take infinite volume limit. One can do the calculation and see that the above indeed happens.

1.5. References and Further Reading

- The introduction Section 1.1 was mostly some re-phrasing of things you can read in the introduction of the notes by David Tong [5] and notes by Micheal Fisher [6]. Chapter 8 of [7], chapter 1 of [4], and chapter 1 of [8] are also really nice.
- The definitions in Section 1.2 are elaborated a bit further in chapter 2 of Goldenfeld's book [4].
- Section 1.3 is based on two resources: First chapter of [9], and the third chapter of [5]. The original reference [10] that put forward this view is also very readable.

2. Phases of the Ising Model

The Ising model is the drosophila of statistical mechanics.

— Nigel Goldenfeld

We will discuss the phase structure of the Ising model in some detail here. We will answer it in three main angles. First, a qualitative description of the phase structure. Second, we will compute critical exponents that can be checked against experiments. Third, we will describe the mechanism that drives the system from order to disorder as we change the temperature.

2.1. Qualitative Picture of the Phases

2.1.1. The Effective Field Theory

From experiments, we know that the Ising model lies in the universality class that describes anisotropic magnets, and liquid-vapor transition. With this in mind, we look for an effective field theory that might describe the Ising universality class.

First, we guess an order parameter, or the relevant degrees of freedom. In the magnet experiments, average magnetization dictates order, and in liquid-vapor, the deviation from the critical density plays that role. So a good guess is to take the degree of freedom at long distances to be a scalar field, representing (roughly) the local average magnetization or density.

For the magnets, the coupling between the nearest neighbors is what makes them interesting. To capture this, we propose that the most dominant term that determines the thermodynamics of the system is,

$$S_0 = \int d^d \mathbf{x} \frac{1}{2} \nabla \varphi \cdot \nabla \varphi \quad (2.1)$$

We will choose dimensions of φ such that this term has magnitude one. We will assess the importance of all other terms with respect to this one. To this end, we will enact the second and third steps of RG. For $\zeta > 1$,

$$\mathbf{x} \rightarrow \mathbf{x}' = \zeta^{-1} \mathbf{x}, \quad \varphi(\mathbf{x}) \rightarrow \varphi'(\mathbf{x}') = \zeta^{-\Delta} \varphi(\mathbf{x}) \quad (2.2)$$

We get⁸,

$$S_0 = \int d^d \mathbf{x} \zeta^{d-2+2\Delta} \frac{1}{2} \nabla \varphi \cdot \nabla \varphi \quad (2.3)$$

⁸Not writing primes. Just think of replacing $\mathbf{x}$ with $\zeta \mathbf{x}$ and φ with $\zeta^{-\Delta} \varphi$.

and choose,

$$\Delta = \frac{2-d}{2} \quad (2.4)$$

so that the free term stays at the same magnitude.

Now, we play the effective field theory game. We will write down all possible terms that are allowed by symmetries. The symmetries are the following,

- Rotations in space. This we expect to be an emergent symmetry for a magnet, which comes from the underlying discrete rotational symmetry of the magnet.
- Translations in space. This is again an emergent symmetry from the discrete version in the lattice.
- $\mathbb{Z}_2$ symmetry, when $B = 0$. The Ising model had $\mathbb{Z}_2$ symmetry when external magnetic field was zero. Similarly for the liquid-vapor system at $P = P_c$, we had $\rho - \rho_c \rightarrow -(\rho - \rho_c)$ symmetry.

We write down all possible terms that has the above symmetries. Since we have two knobs we can turn, namely temperature and external magnetic field, we can afford to have at most two relevant interactions. If we find more than two relevant interactions, we failed to find a natural theory and would have to restart.

Linear terms

$$S_1 = \int d^d \mathbf{x} g_{1,0} \varphi \xrightarrow{\text{scaling}} \int d^d \mathbf{x} \zeta^{\frac{d+2}{2}} g_{1,0} \varphi \quad (2.5)$$

$$g_1(\zeta) = \zeta^{\frac{d+2}{2}} g_{1,0}$$

This is relevant for any d . But this term has a natural identification with the tunable variable $-B$. If φ is the local average magnetization, one would find the mean magnetization by,

$$m = \langle \varphi(\mathbf{x}) \rangle = \left[\frac{1}{\text{vol}} \frac{\delta}{\delta B(\mathbf{x})} \ln \left(\int \mathcal{D}\varphi e^{-S_0 + \int d^d \mathbf{x} B(\mathbf{x}) \varphi(\mathbf{x})} \right) \right]_{B=0} \quad (2.6)$$

This is essentially the same as computing m from the microscopic Ising model. So we identify g_1 with the external magnetic field.

Now we can add derivatives to the linear term. We must add even number of derivatives otherwise we will violate rotational symmetry in space. The first such term is with two derivatives. But this term is a total derivative, unless we write $g(x) \nabla^2 \varphi$. But this is not good, as it introduces new degrees of freedom. We don't want that at the moment. So

by the same logic, every higher derivative term on a linear φ is a boundary term that we discard.

Quadratic terms

The simplest is,

$$S_2 = \int d^d \mathbf{x} \mu_0^2 \varphi^2 \xrightarrow{\text{scaling}} \int d^d \mathbf{x} \zeta^{d+2\frac{2-d}{2}} \mu_0^2 \varphi^2$$

$$\mu^2(\zeta) = \zeta^2 \mu_0^2 \quad (2.7)$$

This term is relevant for all d . This is okay, since we have one more tunable parameter – the temperature. If we find one more relevant term, we have to restart.

Let us add derivatives. Adding two is the same as our free term, so we must add four. Then, the scaling will be ζ^{-2} , and is irrelevant. Adding more derivatives makes it more and more irrelevant.

Cubic terms

$$S_3 = \int d^d \mathbf{x} g_{3,0} \varphi^3 \xrightarrow{\text{scaling}} \int d^d \mathbf{x} \zeta^{d+3\frac{2-d}{2}} g_{3,0} \varphi^3$$

$$g_3(\zeta) = \zeta^{\frac{6-d}{2}} g_{3,0} \quad (2.8)$$

This term is relevant as long as $d < 6$. Seems like we have failed to describe the Ising model with a scalar order parameter. There is actually a catch. This is that the cubic term that we wrote above is redundant, meaning it contains no new physics. We will be patient for now, and explain this once we have exhausted all relevant terms.

Adding more derivatives to the cubic term will make it irrelevant, which can be checked trivially.

Quartic terms

$$S_4 = \int d^d \mathbf{x} g_{4,0} \varphi^4 \xrightarrow{\text{scaling}} \int d^d \mathbf{x} \zeta^{d+4\frac{2-d}{2}} g_{4,0} \varphi^4$$

$$g_4(\zeta) = \zeta^{4-d} g_{4,0} \quad (2.9)$$

This term is irrelevant for $d > 4$, marginal for $d = 4$, and relevant for $d \leq 3$. Therefore, we already see that it is impossible to describe the Ising model with this simple model in dimensions $d \leq 3$. That's a bummer, since all of the interesting experiments are done for three and two dimensional magnets.

Adding more derivatives as before makes the term irrelevant for $d \geq 3$.

Fifth and Sixth order terms

It is easy to check that the φ^5 term is irrelevant for $d = 4$, but is relevant for $d \leq 3$. The φ^6 term is marginal for $d = 3$. As we see, this simple scalar field model is inadequate to describe the Ising model in $d \leq 3$. We will first understand the $d = 4$ Ising model using this simple scalar field theory. Then at the end, we will do something desperate to infer something useful about the $d = 3$ model from whatever we have, and try to make some quantitative predictions. $d = 2$ requires more creativity, and turns out to be actually exactly solvable due to the magic of two dimensional conformal field theories. This perhaps will be discussed in a later chapter.

Before moving on to the analysis of the effective field theory, let us tie the loose end that we left in the φ^3 term. Since we have the φ^4 term, and lack of $\mathbb{Z}_2$ symmetry when $B \neq 0$, we are free to shift the field $\varphi \rightarrow \varphi + c$. Then, we can absorb the φ^3 term to $c\varphi^3 \subset (\varphi + c)^4$ term. For this reason, the cubic term has no additional physics other than what we can describe via the linear, quadratic and quartic terms in $d = 4$.

2.1.2. The Phase Structure of the $d = 4$ Theory

Our effective field theory for the $d = 4$ Ising model is,

$$S = \int d^4x \left[\frac{1}{2} \nabla \varphi \cdot \nabla \varphi + \frac{1}{2} \mu_0^2 \varphi^2 + \frac{\lambda_0}{4!} \varphi^4 - B \varphi \right] \quad (2.10)$$

As explained before, we have two relevant parameters, one of them we identified with B . The other one is μ_0^2 . This parameter therefore must be tunable by changing the temperature. λ_0 is marginal, and we assume it is positive, as a negative λ_0 will result in no ground states for the theory. Due to thermal fluctuations, λ_0 can become relevant or irrelevant at long distances, which we will see in the next section.

From the general analysis, we know that at the critical point, we can have no length scales in the problem, and therefore all relevant parameters must be zero. To get to the critical point, we tune $B \rightarrow 0$, and $T \rightarrow T_c$. This imply,

$$\lim_{T \rightarrow T_c} \mu_0^2 = 0 \quad (2.11)$$

Around T_c , we can Taylor expand μ_0^2 , and it must have the form,

$$\mu_0^2 \sim \pm(T - T_c) + \mathcal{O}((T - T_c)^2) \quad (2.12)$$

We will pick the positive sign for now, and see what that gives us. With this choice, we have,

$$\begin{aligned}\mu_0^2 &< 0 \text{ for } T < T_c, \\ \mu_0^2 &> 0 \text{ for } T > T_c\end{aligned}\tag{2.13}$$

Now we ask: does this model describe the Ising model at least qualitatively? How would we go about answering this question? Well, the simple model should at least tell us that there is order below T_c . To check if this is true, we have to compute the average magnetization $m = \langle \varphi(\mathbf{x}) \rangle$. Computing the partition function exactly is impossible due to the φ^4 term. What we will do is to approximate the partition function by its saddle point.

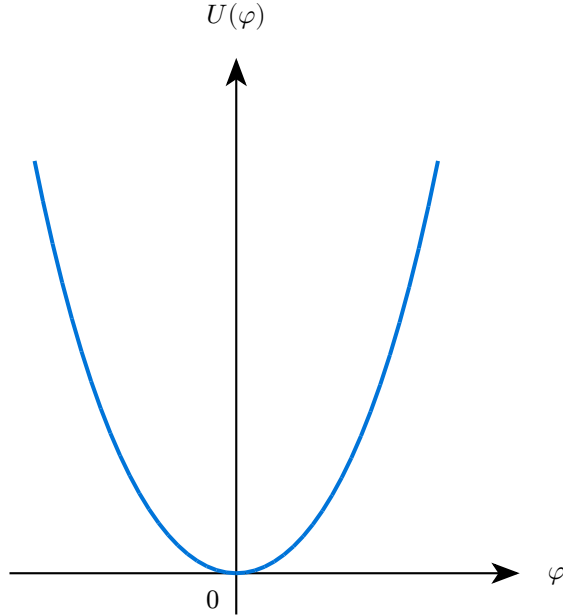
Let us first consider $T > T_c$. Writing the action as,

$$S = \int d^4\mathbf{x} \left[\frac{1}{2} \nabla \varphi \cdot \nabla \varphi + U(\varphi(\mathbf{x})) - B\varphi(\mathbf{x}) \right] \tag{2.14}$$

where,

$$U(\varphi(\mathbf{x})) = \frac{1}{2} \mu_0^2 \varphi^2 + \frac{\lambda_0}{4!} \varphi^4 \tag{2.15}$$

plotting U as a function of constant φ , we have,

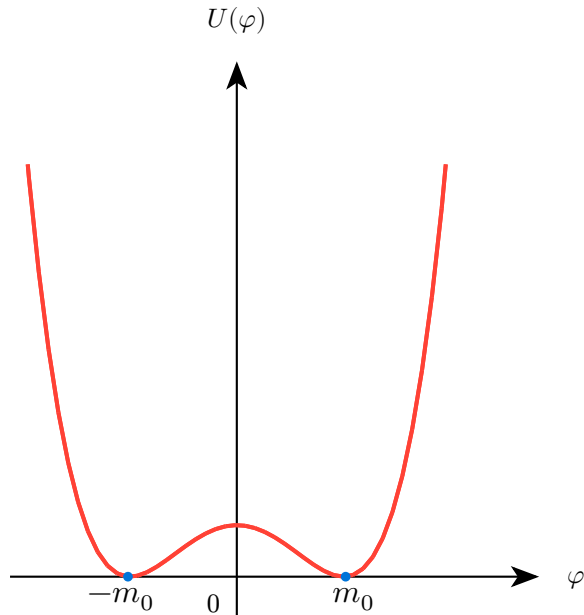


We see that the global minimum for U is at $\varphi = 0$. As a function of $\mathbf{x}$, $U(\varphi(\mathbf{x}))$ must go to zero as $|\mathbf{x}| \rightarrow \infty$ to have a finite action. Infinite action configurations are suppressed in the partition function. For U , the only option is $\varphi(|\mathbf{x}| \rightarrow \infty) \rightarrow 0$. To minimize the gradient term in the action, we can simply choose $\varphi(\mathbf{x}) = 0$. This is the most dominant contribution to the partition function in the thermodynamic limit.

This tells us that,

$$m = \langle \varphi(\mathbf{x}) \rangle = 0, \text{ for } T > T_c \quad (2.16)$$

Now, let us turn to $T < T_c$, implying $\mu_0^2 < 0$. Then, U as a function of constant φ looks like,



As before, U as a function of $\mathbf{x}$ should go to zero as $|\mathbf{x}| \rightarrow \infty$ to have finite action. There are several choices now, $\varphi(|\mathbf{x}| \rightarrow \infty) \rightarrow \pm m_0 = \pm \sqrt{\frac{6|\mu_0^2|}{\lambda_0}}$. To minimize the gradient, we can choose $\varphi(\mathbf{x}) = \pm m_0$. There are other choices, for example $\varphi(x^1 \rightarrow -\infty) \rightarrow -m_0$, and $\varphi(x^1 \rightarrow \infty) \rightarrow m_0$. This will look like a “wall” that direction that separates the $+m_0$ region from the $-m_0$ region. This configuration is called a domain wall, and it will look like the picture below,

the picture (2.17)

For the domain wall, we cannot set the gradient term to zero. There will be a gradient contribution to the action along the three dimensional hyperplane. So the action cost will be proportional to the area of the three dimensional hyperplane. In the thermodynamic limit, this contribution is suppressed as compared to the simple $\varphi(\mathbf{x}) = \pm m_0$ contribution⁹.

Which of the two minima should we choose? In the absence of the magnetic field, both choices are equally good. Now, let us imagine a small but positive constant magnetic field in the game. Then, the action gets a contribution from the $B\varphi$ term. The action evaluated at the minima is,

⁹We will revisit the domain walls in a later section, when we talk about the mechanism that drives order to disorder.

$$\begin{aligned}
S(\pm m_0) &= \text{Vol} \left[-\frac{1}{2} |\mu_0^2| m_0^2 + \frac{\lambda_0}{4!} m_0^4 \mp B m_0 \right] \\
&= \text{Vol} \left[-\frac{3}{2} \frac{|\mu_0^2|^2}{\lambda_0} \mp B \sqrt{\frac{6|\mu_0^2|}{\lambda_0}} \right]
\end{aligned} \tag{2.18}$$

The lower value for the action for $B > 0$ is given by the $+m_0$ choice. This is what will dominate in the thermodynamic limit. Therefore,

$$m = \langle \varphi(\mathbf{x}) \rangle = \lim_{B \rightarrow 0^\pm} \lim_{\text{Vol} \rightarrow \infty} \frac{1}{\text{Vol}} \frac{\partial}{\partial B} \ln Z = \pm m_0 \tag{2.19}$$

This is indeed what we see in experiments, order below T_c . The system picks one of the two saddles.

A question that we can now ask is if such a state is stable. In order to flip the alignment of the entire magnet, a small closed domain should form, and then it should grow to the entire volume. But this costs infinite energy, and therefore either choice of the ground state is stable in the absence of magnetic field.

2.1.3. Spontaneous Breaking of Symmetry

The effective field theory we started with, when $B = 0$ had the symmetry group $\mathbb{Z}_2 \times \text{SO}(4) \times \mathbb{R}^4$. When $T > T_c$, the equilibrium state was $\varphi(\mathbf{x}) = 0$, which also has the same set of symmetries. For $T < T_c$, the state $\varphi(\mathbf{x}) = \pm m_0$ only has the symmetries $\text{SO}(4) \times \mathbb{R}^4$. The ground state changed its symmetry group as we move from $T > T_c$ to $T < T_c$. This is called spontaneous symmetry breaking. It is this change in symmetry class that in some sense is causing the violent behavior at the critical point.

There is one interesting subtlety here. Usually in physics, we are careless about the order in which we take limits. However for $T < T_c$,

$$\lim_{B \rightarrow 0^\pm} \lim_{\text{Vol} \rightarrow \infty} \frac{1}{\text{Vol}} \frac{\partial}{\partial B} \ln Z = \pm m_0 \tag{2.20}$$

but,

$$\lim_{\text{Vol} \rightarrow \infty} \lim_{B \rightarrow 0^\pm} \frac{1}{\text{Vol}} \frac{\partial}{\partial B} \ln Z = 0 \tag{2.21}$$

To see the second relation, notice that,

$$\langle \varphi(\mathbf{x}) \rangle_{B=0} = \int \mathcal{D}\varphi \varphi(\mathbf{x}) e^{-\int d^4\mathbf{x} \frac{1}{2} \nabla\varphi \cdot \nabla\varphi - \frac{1}{2} |\mu_0^2| \varphi^2 + \frac{\lambda_0}{4!} \varphi^4} = 0 \tag{2.22}$$

due to the fact that the integrand is odd. What is going on? The idea is that for $B = 0$, there are two saddle points for the free energy, both equally likely. So, it becomes impossible to describe the ground state with a pure state, like $\varphi = m_0$. Instead we are forced to resort to the density matrix description, with density matrix (adopting quantum notation, but it has nothing to do with quantum mechanics),

$$\rho = \frac{1}{2}(|+\rangle\langle+| + |-\rangle\langle-|) \quad (2.23)$$

Recall that the averages in statistical mechanics is over an ensemble. Even for $B = 0$ below T_c , a single instance from the ensemble will have either $+m_0$ or $-m_0$ uniformly. But as an ensemble, it will have equal number of each, and therefore the average value of magnetization is going to be zero. In the real world, magnetic fields have fluctuations, and therefore never zero. So in the approximate infinite volume limit, we get a fixed orientation for a magnet when cooled.

Another interesting consequence of spontaneous symmetry breaking is the failure of ergodic theorem. Ergodic theorem states the following: the ensemble average is equal to time average of an observable in one instance of the ensemble. The idea behind is that the system will explore all configurations allowed given enough time. As explained above, for $B = 0$, infinite volume model below T_c , the ensemble average is zero. But if we pick one instance from the ensemble, it will have all spins aligned. This is the only configuration that is allowed, and therefore a time average of average magnetization on this instance will not be zero.

Since the average magnetization at $B = 0$ has this subtlety, it is convenient to define another quantity as the order parameter (the quantity that distinguishes the phases). This is,

$$\lim_{|\mathbf{x}-\mathbf{x}'| \rightarrow \infty} \langle \varphi(\mathbf{x})\varphi(\mathbf{x}') \rangle \quad (2.24)$$

This quantity will be zero when $T > T_c$, and m_0^2 for $T < T_c$ even when $B = 0$, as one can check very easily.

need discussion of the phase plots.

2.2. The Mechanism of Phase Transition

Why does the phase go from order to disorder as we move up in temperature across T_c ? We will try to answer this question in this section. Let us put ourselves below T_c , with zero external field. As we explained before, the least action state is just constant $\varphi(\mathbf{x}) = m_0$. We also discussed another possible solution, which say along the x^1 direction changes value from $+m_0$ to $-m_0$ at $x^1 = \pm\infty$. This solution – called a domain wall – has very large

contribution to the action. Let us nonetheless find the domain wall solution. It satisfies the equation of motion,

$$-\nabla^2\varphi - |\mu_0^2|\varphi + \frac{\lambda}{3!}\varphi^3 = 0 \quad (2.25)$$

we take the wall to be along the x^1 direction. Then we have to solve the ode,

$$-\frac{d^2\varphi}{dx^{12}} - |\mu_0^2|\varphi + \frac{\lambda}{3!}\varphi^3 = 0 \quad (2.26)$$

with the boundary condition mentioned above. The solution turns out to be,

$$\varphi(x^1) = m_0 \tanh\left(\frac{x^1 - X^1}{W}\right), \quad W = \sqrt{\frac{2}{|\mu_0^2|}} \quad (2.27)$$

X^1 will be called the center and W the width of the domain wall. We can plug the solution into the action and find the contribution to the action due to one domain wall. We can estimate it in the following way: the contribution from the domain wall is coming from the gradient term, as it changes from $+$ to $-$. This change is,

$$\frac{d\varphi}{dx^1} \sim \frac{2m_0}{W} \quad (2.28)$$

then,

$$S_{\text{wall}} \sim L^{d-1}W \times \left(\frac{m_0}{W}\right)^2 = L^{d-1}m_0^2\sqrt{|\mu_0^2|} \quad (2.29)$$

The constant solution had free energy that scaled like L^d , which dominates the saddle and therefore is the ground state below T_c .

The domain wall has one other contribution to the partition function, which usually is referred to as the entropic contribution. If we have a domain wall of boundary area A , then there are exponentially large number of ways to construct one. As an example that we can visualize, let us take two dimensions. The boundary area of a domain now is just the perimeter. Imagine dividing the whole space into boxes of side length equal to cut off ε . Imagine constructing a closed curve of perimeter A by joining lines between centers of the boxes. As A increase, the number of ways of constructing this perimeter increases exponentially,

$$\# \text{ of ways} \sim e^{\alpha \frac{A}{\varepsilon^{d-1}}} \quad (2.30)$$

for some α . Therefore, the contribution to the action due to one domain wall with area A is,

$$Z_{\text{wall}} \sim e^{\alpha \frac{A}{\varepsilon^{d-1}} - A m_0^2 \sqrt{|\mu_0^2|}} \quad (2.31)$$

We will call $\sigma = m_0^2 \sqrt{|\mu_0^2|}$ the tension of the domain wall. This also depends on ε . Recall that μ_0^2 has dimensions of inverse length squared. Therefore, at the parameter μ_0^2 defined at the cut-off scale naturally is written as $\mu_0^2 = \frac{\kappa}{\varepsilon^2}$, with κ of order 1. In d dimensions, the λ_0 also have dimensions, and we can see by a bit of analysis that $m_0 \sim \varepsilon^{\frac{2-d}{2}}$. Overall, the string tension σ looks like,

$$\sigma(T) = \frac{\tilde{\sigma}(T)}{\varepsilon^{d-1}}, \quad \text{with } \tilde{\sigma} \in \mathbb{R} \quad (2.32)$$

So the total action in the presence of one domain wall is,

$$S \sim f \frac{V}{\varepsilon^d} + (\tilde{\sigma}(T) - \alpha) \frac{A}{\varepsilon^{d-1}} \quad (2.33)$$

The fV is the usual contribution from the constant solution. The wall has extra contribution only at the surface area as explained before. For T significantly small compared to T_c , from the presence of μ_0 in σ , we can infer that $\tilde{\sigma}$ will be typically large compared to the constant α . So, the action with one wall is going to be smaller than the usual no wall case. Therefore the probabilities,

$$\frac{p_{\text{wall}}}{p_{\text{no-wall}}} = \frac{e^{-f \frac{V}{\varepsilon^d} + |(\tilde{\sigma}(T) - \alpha)| \frac{A}{\varepsilon^{d-1}}}}{e^{-f \frac{V}{\varepsilon^d}}} > 1 \quad (2.34)$$

So the ground state is unlikely to have large domain walls. But, as we increase temperature and get close to T_c , $\sigma \rightarrow 0$, which means it becomes more and more likely to have domain walls. As we get to T_c , domain walls proliferate, meaning we will have domain walls with arbitrary sizes and shapes, everywhere on the space. This essentially destroys the order!

We have almost exhausted the qualitative features of the scalar field model for the Ising universality class. It is now time to compute things...

2.3. Critical Exponents

2.3.1. RG flow in d – Dimensions

We have left one coupling – namely the φ^4 coupling – hanging. Naive scaling tells us that this coupling is important at all scales. But importantly, we have ignored the contribution to the coupling coming from integrating out the high momentum modes. This can

ultimately make this coupling relevant or irrelevant at low energies. We will figure this out now.

Our effective action is,

$$\begin{aligned}
S &= \int d^d \mathbf{x} \left(\frac{1}{2} (\nabla \varphi)^2 + \frac{1}{2} \mu_0^2 \varphi^2 + \frac{\lambda_0}{4!} \varphi^4 - B \varphi \right) \\
&= -B \varphi(\mathbf{k} = 0) + \int \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{2} \varphi(\mathbf{k}) (\mathbf{k}^2 + \mu_0^2) \varphi(-\mathbf{k}) \\
&\quad + \frac{\lambda_0}{4!} \int_{\mathbf{k}_{1,2,3,4}} (2\pi)^4 \delta^4(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 + \mathbf{k}_4) \varphi(\mathbf{k}_1) \varphi(\mathbf{k}_2) \varphi(\mathbf{k}_3) \varphi(\mathbf{k}_4)
\end{aligned} \tag{2.35}$$

We will now decompose the field into low and high momentum modes, denote by,

$$\varphi(\mathbf{k}) = \varphi_+(\mathbf{k}) + \varphi_-(\mathbf{k}) \tag{2.36}$$

Note that the linear in φ term is unaffected by the decomposition, so we will simply work in the $B = 0$ case. Under this decomposition, the action splits into parts, denote by,

$$S = S_{+,0} + S_- + S_{\text{int}} \tag{2.37}$$

with,

$$\begin{aligned}
S_- &= \int_{\mathbf{k}} \frac{1}{2} \varphi_-(\mathbf{k}) (\mathbf{k}^2 + \mu_0^2) \varphi_-(-\mathbf{k}) \\
&\quad + \frac{\lambda_0}{4!} \int_{\mathbf{k}_{1,2,3,4}} (2\pi)^4 \delta^4(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 + \mathbf{k}_4) \varphi_-(\mathbf{k}_1) \varphi_-(\mathbf{k}_2) \varphi_-(\mathbf{k}_3) \varphi_-(\mathbf{k}_4)
\end{aligned} \tag{2.38}$$

$$S_{+,0} = \int_{\mathbf{k}} \frac{1}{2} \varphi_+(\mathbf{k}) (\mathbf{k}^2 + \mu_0^2) \varphi_+(-\mathbf{k}) \tag{2.39}$$

and,

$$\begin{aligned}
S_{\text{int}} &= \int_{\mathbf{k}} \varphi_+(\mathbf{k}) (\mathbf{k}^2 + \mu_0^2) \varphi_-(-\mathbf{k}) \\
&\quad + \frac{\lambda_0}{4!} \int_{\mathbf{k}_{1,2,3,4}} (2\pi)^4 \delta^4(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 + \mathbf{k}_4) [\varphi_+^4 + 4\varphi_+^3 \varphi_- + 6\varphi_+^2 \varphi_-^2 + 4\varphi_+ \varphi_-^3]
\end{aligned} \tag{2.40}$$

the first term is zero since φ_- and φ_+ have mutually exclusive ranges of momenta allowed.

Recall, we defined the new effective action with a lower cut-off after integrating out the $+$ fields as,

$$e^{-\tilde{S}[\varphi_-]} = \int \mathcal{D}\varphi_+ e^{-S} = e^{-S_-} \langle e^{-S_{\text{int}}} \rangle_+ \quad (2.41)$$

where the expectation value above is with respect to the free action $S_{+,0}$ for the $+$ fields. Taking a log,

$$\tilde{S}[\varphi_-] = S_-[\varphi_-] - \ln \langle e^{-S_{\text{int}}} \rangle_+ \quad (2.42)$$

We will now do the following. We assume that λ_0 is small, so that we can do perturbation theory. Then,

$$\begin{aligned} \ln \langle e^{-S_{\text{int}}} \rangle_+ &= \ln \left\langle 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} S_{\text{int}}^n \right\rangle_+ \\ &= -\langle S_{\text{int}} \rangle_+ + \frac{1}{2!} \left[\langle S_{\text{int}}^2 \rangle_+ - \langle S_{\text{int}} \rangle_+^2 \right] + \dots \end{aligned} \quad (2.43)$$

It turns out that each term in the series will be a connected correlation function, or cumulant. That is,

$$\ln \langle e^{-S_{\text{int}}} \rangle_+ = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \langle S_{\text{int}}^n \rangle_{+,c} \quad (2.44)$$

the subscript c denoting connected correlators. The canonical way to arrange these connected correlators is using Feynman diagrams. Here is how we will do it. Notice that terms in S_{int} have φ_+ and φ_- . The slow modes φ_- are not being integrated over, but all the fast modes are. We will therefore think of the φ_- as external legs of a Feynman diagram, and φ_+ will always be internal lines. We will denote φ_- with a solid line and φ_+ with a dotted line. From each term in the interaction, we will have a verity of vertices, with some dotted and some solid lines. A diagram with say two solid lines will be an additional contribution to the quadratic in φ_- terms in S_- , and so on. Let us do this order by order in λ_0 .

Linear in λ_0

We need to compute $\langle S_{\text{int}} \rangle_+$. Since the free action of choice for the $+$ fields is Gaussian, we have $\langle \varphi_+ \rangle_+ = 0$. In the same vein, all the terms with odd powers of φ_+ vanish. The φ_+^4 term will have no external lines, therefore will be just a number, albeit divergent. Since we are interested in the correction to the φ_- action, the only term that is of interest to us

will be the $\varphi_-^2 \varphi_+^2$, which will correct the Gaussian part of the φ_- action. We can compute it directly,

$$\langle S_{\text{int}} \rangle_+ = \frac{\lambda_0}{4} \int_{\mathbf{k}_{1,2,3,4}} (2\pi)^4 \delta^4 \left(\sum_{i=1}^4 \mathbf{k}_i \right) \varphi_-(\mathbf{k}_1) \varphi_-(\mathbf{k}_2) \langle \varphi_+(\mathbf{k}_3) \varphi_+(\mathbf{k}_4) \rangle_+ \quad (2.45)$$

The two point function is easy as the free action is Gaussian,

$$\langle \varphi_+(\mathbf{k}_3) \varphi_+(\mathbf{k}_4) \rangle_+ = (2\pi)^4 \delta^4(\mathbf{k}_4 + \mathbf{k}_3) \frac{1}{\mathbf{k}_3^2 + \mu_0^2} \quad (2.46)$$

Plugging this in, we get,

$$\langle S_{\text{int}} \rangle_+ = \frac{\lambda_0}{4} \int_{\mathbf{k}_1, \mathbf{k}_2} (2\pi)^4 \delta^4(\mathbf{k}_1 + \mathbf{k}_2) \varphi_-(\mathbf{k}_1) \varphi_-(\mathbf{k}_2) \int_{\frac{\Lambda}{\zeta}}^{\Lambda} \frac{d^4 \mathbf{k}}{(2\pi)^4} \frac{1}{\mathbf{k}^2 + \mu_0^2} \quad (2.47)$$

What happened above can be phrased in the following way. We had a four point vertex with momentum conserved. The expectation value collapsed the $+$ legs to a loop. Diagrammatically,

diagram here

In general, this is what will happen. So we can write down schematics for Feynman diagrams in general.

1. Draw a connected diagram with N external solid lines to find the contribution to the φ_-^N term, with all possible vertices that we have from S_{int} .
2. Link up all internal dotted lines. Each internal line is a propagator $\langle \varphi_+(\mathbf{k}) \varphi_+(\mathbf{k}') \rangle_+$.
3. Conserve momentum at each vertex.
4. Perform the internal loop integrals.

At the first order in λ_0 , this is the only non-trivial term. This new term is nothing but $\delta\mu^2$. Now we do the scaling steps of RG as well, to arrive at,

$$\mu^2(\zeta) = \zeta^2 \left(\mu_0^2 + \frac{\lambda_0}{4} I_d(\zeta) \right), \quad \lambda(\zeta) = \zeta^{4-d} \lambda_0 \quad (2.48)$$

where,

$$I_d(\zeta) = \int_{\frac{\Lambda}{\zeta}}^{\Lambda} \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{\mathbf{k}^2 + \mu_0^2} \quad (2.49)$$

Previously, we claimed that to hit the critical point, we should tune the parameter μ_0^2 to zero by tuning the external temperature. But we see that this is false. Even if we had μ_0^2 to be zero to start with, λ_0 induces a μ^2 term. So in experiments, when we tune T to T_c , we must be tuning the linear combination $\mu_0^2 + \frac{\lambda_0}{4} I_d(\zeta)$ to zero. So the relevant direction in our system is not the simple μ_0^2 , but this linear combination. The λ_0 direction in theory space remain marginal. Both of these statements will be corrected in general, as we go higher up in perturbation theory.

Quadratic in λ_0

We can either start from S_{int}^2 and figure out which of them are interesting to us, or simply draw Feynman diagrams. We will do the latter. At this order, we are able to draw diagrams with four external solid lines, as we are squaring the two external line term. There are several possibilities, one of which is shown below,

diagram here

This is simply the square of the $\mathcal{O}(\lambda_0)$ term that we computed earlier, therefore is equal to $\langle S_{\text{int}} \rangle_+^2$. These terms are cancelled automatically, and we only need to look at connected correlators. Therefore, the only diagram at this order that goes into $\langle S_{\text{int}}^2 \rangle_{+,c}$ with four external solid lines is this,

diagram here

The diagram evaluates to,

$$\frac{\lambda_0^2}{16} \int_0^{\frac{\Lambda}{\zeta}} \prod_{n=1}^4 \left[\frac{d^d \mathbf{k}_n}{(2\pi)^d} \varphi_-(\mathbf{k}_n) \right] (2\pi)^d \delta^d \left(\sum_{i=1}^4 \mathbf{k}_i \right) f(\mathbf{k}_1 + \mathbf{k}_2) \quad (2.50)$$

where the loop integral f is,

$$f(\mathbf{k}_1 + \mathbf{k}_2) = \int_{\frac{\Lambda}{\zeta}}^{\Lambda} \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{\mathbf{k}^2 + \mu_0^2} \frac{1}{(\mathbf{k} + \mathbf{k}_1 + \mathbf{k}_2)^2 + \mu_0^2} \quad (2.51)$$

Since the momenta $\mathbf{k}_{1,2}$ are bounded above at $\frac{\Lambda}{\zeta}$, we can safely approximate,

$$f(\mathbf{k}_1 + \mathbf{k}_2) \equiv f_d(\zeta) \approx \int_{\frac{\Lambda}{\zeta}}^{\Lambda} \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{(\mathbf{k}^2 + \mu_0^2)^2} \quad (2.52)$$

We therefore have obtained a correction to the φ^4 coupling. Doing the other steps of RG as well, we get,

$$\lambda(\zeta) = \zeta^{4-d} \left(\lambda_0 - \frac{\lambda_0^2}{32} f_d(\zeta) \right) \quad (2.53)$$

the negative sign and a factor of $\frac{1}{2}$ coming from the cumulant expansion in Equation (2.44).

There is actually one more diagram with four external legs, as shown below. But this is zero since the vertex with three dotted and one undotted line cannot satisfy momentum conservation. The loop of the dotted line imply the momentum of solid line and dotted internal line are equal, which is impossible.

diagram here.

2.3.2. Beta Functions in $d = 4$ Dimensions

At one loop order, we have,

$$\mu^2(\zeta) = \zeta^2 \left(\mu_0^2 + \frac{\lambda_0}{4} I_d(\zeta) \right), \quad \lambda(\zeta) = \zeta^{4-d} \left(\lambda_0 - \frac{\lambda_0^2}{32} f_d(\zeta) \right) \quad (2.54)$$

We imagine ζ is infinitesimally close to 1, so that we can interpret the above as a differential equation. We can then approximate,

$$I_d(\zeta) = \int_{\frac{\Lambda}{\zeta}}^{\Lambda} \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{\mathbf{k}^2 + \mu_0^2} \approx \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})(2\pi)^d} \left(1 - \frac{1}{\zeta} \right) \frac{\Lambda^d}{\Lambda^2 + \mu_0^2} \quad (2.55)$$

and,

$$f_d(\zeta) \approx \int_{\frac{\Lambda}{\zeta}}^{\Lambda} \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{(\mathbf{k}^2 + \mu_0^2)^2} \approx \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})(2\pi)^d} \left(1 - \frac{1}{\zeta} \right) \frac{\Lambda^d}{(\Lambda^2 + \mu_0^2)^2} \quad (2.56)$$

writing $\zeta = 1 + s$, we can find for $d = 4$,

$$\frac{d\mu^2}{ds} = 2\mu^2 + \frac{\lambda}{32\pi^2} \frac{\Lambda^4}{\Lambda^2 + \mu^2}, \quad \frac{d\lambda}{ds} = -\frac{\lambda^2}{256\pi^2} \frac{\Lambda^4}{(\Lambda^2 + \mu^2)^2} \quad (2.57)$$

as we increase s (increase ζ), μ^2 increases, but λ decreases. That is, in the infrared, λ is an irrelevant coupling.

We can now study the renormalization group flow in the space of couplings. This is done in the usual way, by finding fixed points of the flow equations. Staring the above equations, we see that $\mu^2 = 0, \lambda = 0$ is a fixed point, where the couplings ceases to change. This is also the only fixed point. For the beta function for λ to vanish, we must have $\lambda = 0$, and therefore $\mu^2 = 0$ is necessary for the beta function for μ^2 to vanish.

We can plot the flow in the two dimensional plane of μ^2 vs λ , as parameterized by s . Let us normalize things to simplify. μ^2 has dimensions of L^{-2} , so let us write $\mu^2 = \Lambda^2 y$, and $\frac{\lambda}{32\pi^2} = x$. Then,

$$\frac{dx}{ds} = -\frac{1}{8} \left(\frac{x}{1+y} \right)^2, \quad \frac{dy}{ds} = 2y + \frac{x}{1+y} \quad (2.58)$$

We will not bother solving it everywhere, but solve it around the fixed point, which can be done by linearizing around it. Around $(x, y) = (0, 0)$, we have,

$$\frac{dx}{ds} = 0, \quad \frac{dy}{ds} = 2y + x \Rightarrow \frac{d}{ds} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad (2.59)$$

we can diagonalize, and get eigen values and eigen vectors to be,

$$\begin{pmatrix} 2 \\ -1 \end{pmatrix}, 0 \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}, 2 \quad (2.60)$$

In the $x - y$ plane, around $(0, 0)$, no flow occurs in the $2\hat{x} - \hat{y}$ direction, while diverging flow along $\pm\hat{y}$ direction. From the second order correction, the flow along $2\hat{x} - \hat{y}$ direction can be inferred to be directed towards $(0, 0)$. We can plot the flow as shown below,

plot

A system not typically at critical point has some arbitrary value of the parameters x and y (and the infinitely many irrelevant parameters too). RG flow takes it to either of the phases, depending upon where the starting values lie relative to the red line shown above. That line separates the ordered and disordered phase. Then, we tune the external parameters, and that moves around the starting point of the microscopic theory. If we manage to hit the red line, RG flow takes us to the critical point in the IR, and we observe critical phenomena. This is in some sense why universality is true. The microscopic starting values of the couplings could be anything, but since there is only one critical point, if we tune external parameters to hit the red line anywhere, we are guaranteed to converge to the critical point, no matter what.

2.3.3. Bete Functions in $d = 3.99999\dots$ Dimensions

As mentioned before, the effective field theory that we wrote down is inadequate to describe the $d = 3$ Ising model, as we have too many relevant parameters than external knobs that we can turn. However, we will now do something desperate, and try to extract some qualitative features of the $d = 3$ model, from whatever we have so far.

We take the RG flow equations, and analytically continue this to non-integer dimensions. This is slightly crazy, but as we will see will give us qualitative (an in the next section,

some quantitative) idea about the $d = 3$ model. First, we find beta functions in general d – dimensions, assuming $d \geq 4$. This is done just like before. Write $\zeta = 1 + s$, with small s . Then,

$$\frac{d\mu^2}{ds} = 2\mu^2 + \frac{\lambda}{4} \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})(2\pi)^d} \frac{\Lambda^d}{\Lambda^2 + \mu^2}, \quad \frac{d\lambda}{ds} = (4-d)\lambda - \frac{\lambda^2}{32} \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})(2\pi)^d} \frac{\Lambda^d}{(\Lambda^2 + \mu^2)^2} \quad (2.61)$$

Let us define $d = 4 - \varepsilon$. As before, $\mu^2 = \Lambda^2 y$, and $\frac{\lambda}{4} \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})(2\pi)^d} = \Lambda^\varepsilon x$. Then,

$$\frac{dx}{ds} = \varepsilon x - \frac{1}{8} \left(\frac{x}{1+y} \right)^2, \quad \frac{dy}{ds} = 2y + \frac{x}{1+y} \quad (2.62)$$

Now, we have an additional fixed point for the RG flow, other than the $(x, y) = (0, 0)$ point. This is known as the *Wilson-Fisher* fixed point,

$$(x_\star, y_\star) = \left(8\varepsilon(1+y_\star)^2, -\frac{x_\star}{2(1+y_\star)} \right) \quad (2.63)$$

We will expand in small ε ,

$$(x_\star, y_\star) \approx (8\varepsilon, -4\varepsilon) \quad (2.64)$$

For $\varepsilon > 0$, both the x and y directions are unstable around $(0, 0)$. What about this new fixed point? We linearize the equations near the fixed point to find the flow near it. Let $x = x_\star + \delta x$ and $y = y_\star + \delta y$. Then

$$\frac{d}{ds} \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = \begin{pmatrix} -\varepsilon & 0 \\ 1 & 2 - 8\varepsilon \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} \quad (2.65)$$

The matrix is the Jacobian,

$$\begin{pmatrix} \frac{\partial X(x,y)}{\partial x} & \frac{\partial X(x,y)}{\partial y} \\ \frac{\partial Y(x,y)}{\partial x} & \frac{\partial Y(x,y)}{\partial y} \end{pmatrix}_{(x_\star, y_\star)} \quad (2.66)$$

with X and Y being the right hand sides of the beta function equations. To solve this linear equation, we diagonalize the Jacobian, and get eigenvalues and eigen vectors as,

$$-\varepsilon, \begin{pmatrix} 1 \\ -\frac{1}{2} \end{pmatrix} \quad 2 - 8\varepsilon, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (2.67)$$

The direction $\delta x - \frac{\delta y}{2}$ is stable, and the δy direction is unstable. Importantly, we only have one unstable direction at the Wilson-Fisher point. That is, we only need to tune one parameter to zero to hit this fixed point!

The flow diagram looks like,

flow diagram

Qualitatively, we expect nothing drastic to happen as we increase ε . Therefore, the critical point that we are hitting by tuning the temperature in $d = 3$ must be the Wilson-Fisher point. This is rather quick. If we recall, at $d = 3$ we had other relevant interactions like φ^6 term. We need to check if such interactions are irrelevant or not. We will do that now, and confirm that they are indeed irrelevant at the WF point. This gives us confidence that the qualitative picture of RG flow stays similar to the above, and therefore $d = 3$ Ising model corresponds to the Wilson-Fisher critical point.

2.3.4. Scaling and Critical Exponents

Now that we have understood the flow diagrams and the different phases of Ising model, it is time to compute critical exponents. In the discussion so far, scaling played a major role in determining the physics of the system. We will see now that using simple scaling arguments, we can derive relationships between the critical exponents.

At the critical point, we have no length scales in the problem. This underlies the fact that at the critical point, we have our correlation function to be a power law. That is,

$$\langle \varphi(\mathbf{x})\varphi(0) \rangle \sim \frac{1}{|\mathbf{x}|^{d-2+\eta}} \quad (2.68)$$

We have parameterized the behavior of the correlation function at the critical point using η . Recall that under scaling, $\mathbf{x} \rightarrow \mathbf{x}' = \zeta^{-1}\mathbf{x}$, we had $\varphi \rightarrow \varphi' = \zeta^{-\Delta_\varphi}\varphi$. To have consistency, we therefore must have,

$$\Delta_\varphi = \frac{2-d-\eta}{2} \quad (2.69)$$

Previously, we simply had $\eta = 0$. But this is only true near a Gaussian fixed point. We will see this soon, and η is one of the critical exponents that we will compute.

If we move slightly away from a critical point, we introduce a new length scale into the problem: the correlation length. It must scale as,

$$\xi \rightarrow \zeta^{-1}\xi \Rightarrow \Delta_\xi = 1 \quad (2.70)$$

Close to the critical point, we have only one relevant direction for the Ising model in zero magnetic field, which we identified with the temperature. Let the scaling of this relevant direction close to the critical point be,

$$t \rightarrow \zeta^{-\Delta_t} t \quad (2.71)$$

This scaling is simple to compute, as we will see in the next section.

The correlation length is supposed to diverge at the critical point. So, very close to critical point, we should have,

$$\xi \sim t^{-\nu}, \quad \text{where } t = \frac{|T - T_c|}{T_c} \quad (2.72)$$

Here, ν is a critical exponent. But ν is related to the scaling of t . From the above relation, we must have,

$$\nu = -\frac{1}{\Delta_t} \quad (2.73)$$

We are going to assume that ξ is the only length scale in the problem close to a critical point. Let us now think about the scaling of the thermodynamic free energy. Intuitively, a correlation length ξ means we have spins correlated in blocks of size ξ^d . Therefore, we expect

$$F_{\text{thermo}} \propto \frac{L^d}{\xi^d} \sim t^{d\nu} \quad (2.74)$$

The heat capacity is then computed from,

$$c \sim \frac{1}{L^d} \frac{\partial^2 F_{\text{thermo}}}{\partial t^2} \sim t^{d\nu-2} \quad (2.75)$$

We see that the critical exponent α is related to ν ,

$$\alpha = 2 - d\nu \quad (2.76)$$

Next, we take up the critical exponent that tell us how the average magnetization scales with temperature. That is,

$$\varphi \sim t^\beta \quad (2.77)$$

For this to be consistent, we must have,

$$\Delta_\varphi = \beta \Delta_t \Rightarrow \beta = \frac{2-d-\eta}{2\Delta_t} \quad (2.78)$$

For the next critical exponent, we must turn on the magnetic field term, $\int B\varphi$. For this term to have the correct scaling dimension, we must have,

$$\Delta_\varphi + \Delta_B = -d \Rightarrow \Delta_B = \frac{-d-2+\eta}{2} \quad (2.79)$$

Now,

$$\chi \sim \left[\frac{\partial \varphi}{\partial B} \right]_{T_c} \sim t^{-\gamma} \quad (2.80)$$

giving us,

$$\gamma = -\frac{\Delta_\varphi - \Delta_B}{\Delta_t} = -\frac{2-\eta}{\Delta_t} \quad (2.81)$$

The last one is,

$$\varphi \sim B^{\frac{1}{\delta}} \quad (2.82)$$

giving us,

$$\Delta_\varphi = \frac{\Delta_B}{\delta} \Rightarrow \delta = \frac{d+2-\eta}{d-2+\eta} \quad (2.83)$$

The scaling has reduced the amount of work that we have. We only have to compute any two of the exponents, and the rest follows. We will choose to compute Δ_t (and therefore ν) and η , and find the rest via the scaling relations.

2.3.5. Finally, Computing the Exponents

Let us start with the $d = 4$ model, where the RG flow equations near the critical point are,

$$\frac{d}{ds} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad (2.84)$$

After diagonalizing the matrix, we can define two new variables $u_1 = 2x - y$ and $u_2 = y$, where the flow equations are,

$$\frac{d}{ds} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad (2.85)$$

The u_1 direction is marginal and u_2 relevant at this order. But we know that beyond linearized, u_1 direction is irrelevant. We identify the u_2 direction as being controlled by the temperature. The solution to RG equations is,

$$u_2(s) = e^{2s}u_2(0) = \zeta^2 u_2(0) \quad (2.86)$$

This is telling us that close to the critical point, u_2 scales as $u_2 \rightarrow \zeta^2 u_2$. If we identify $u_2 \sim t$, then we can infer $\Delta_t = -2$.

We now look at the two point function near the fixed point,

$$\langle \varphi(\mathbf{x}) \varphi(0) \rangle = \int \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{\mathbf{k}^2 + \mu_0^2} e^{i\mathbf{k} \cdot \mathbf{x}} \quad (2.87)$$

At the critical point, we can simply set μ_0^2 to zero, and compute this integral. From dimensional analysis itself, we get that $\eta = 0$. We now have everything we need to proceed. First, we get, $\nu = -\frac{1}{\Delta_t} = \frac{1}{2}$. This gives the rest of the exponents,

$$\alpha = 0, \quad \beta = \frac{1}{2}, \quad \gamma = 1, \quad \delta = 3, \quad \nu = \frac{1}{2}, \quad \eta = 0. \quad (2.88)$$

Let us now move on to the $d = 3.99\dots$ model. Here, the RG flow equations near the critical point are, the last section seem to have the wrong coefficient. This is coming from the incorrect identification for λ . Recall, lambda was defined with a $1/4!$. But when we found the RG running of lambda in the last section, this factor was missed, and this results in wrong eigenvalue. Now we need the correct eigenvalue, not just want the correct sign.

$$\frac{d}{ds} \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = \begin{pmatrix} -\varepsilon & 0 \\ 1 & 2 - \frac{\varepsilon}{3} \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} \quad (2.89)$$

Once again, defining $u_1 = 2\delta x - \delta y$ and $u_2 = \delta y$, we get,

$$\frac{d}{ds} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} -\varepsilon & 0 \\ 0 & 2 - \frac{\varepsilon}{3} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad (2.90)$$

The relevant direction is u_2 , and we have,

$$u_2(\zeta) = e^{s(2-\frac{\varepsilon}{3})} u_2(0) \quad (2.91)$$

giving us, $\Delta_t = \frac{\varepsilon}{3} - 2$. Now, non-zero η appear only if we change the scaling of φ via the field renormalization, $\varphi \rightarrow \sqrt{\gamma} \varphi$, the third step of RG. At the order we are working, we had none. So we simply have $\eta = 0$. We can now compute all the exponents,

$$\alpha = \frac{\varepsilon}{6}, \quad \beta = \frac{1}{2} - \frac{\varepsilon}{6}, \quad \gamma = 1 + \frac{\varepsilon}{6}, \quad \delta = 3 + \varepsilon, \quad \nu = \frac{1}{2} \left(1 + \frac{\varepsilon}{6} \right), \quad \eta = 0. \quad (2.92)$$

What is this good for? We can try just plugging in $\varepsilon = 1$ to get an estimate for the $d = 3$ result, even though this method is not supposed to work there. Here are the results, and a comparison to the experimental values,

	α	β	γ	δ	ν	η
$\varepsilon = 1$ result	0.16	0.33	1.16	4	0.58	0
Expt. result	0.11	0.33	1.24	4.79	0.63	0.04

As we see, the answer are close, even with the dishonest method!

3. Phases of $O(N)$ Models

3.1. Non-Linear Sigma Models

have to write in the initial part after.

Let us denote our field by $\mathbf{n}(x)$. The field comes with a cut-off,

$$\mathbf{n}(x) = \int_0^\Lambda \frac{d^d \mathbf{k}}{(2\pi)^d} e^{i\mathbf{k}\cdot\mathbf{x}} \mathbf{n}(\mathbf{k}) \quad (3.1)$$

We insist that the field is normalized such that,

$$\mathbf{n} \cdot \mathbf{n} = 1 \quad (3.2)$$

The action is,

$$S[\mathbf{n}] = \int d^d x \frac{1}{2g^2} \nabla \mathbf{n} \cdot \nabla \mathbf{n} \quad (3.3)$$

This seems like a simple problem, as the action looks like that of a free field. But this is wrong, due to the constraint on $\mathbf{n}$. We can think of $\mathbf{n}$ as coordinates on a sphere S^{N-1} , and all the interactions are hidden in the constraint. There is one key difference from the RG problems that we did so far. It is the normalization. Thus far, we chose to normalize the field such that the gradient term has unit coefficient. But here, the natural normalization condition is on the magnitude of $\mathbf{n}$, and we measure the strength of interactions with respect to the magnitude of the field vector.

Now, we go on and do our usual RG procedure, and ask what happens to the coupling g^2 . If we integrate out modes in the range $[\frac{\Lambda}{\zeta}, \Lambda]$, the resulting effective action also should have $O(N)$ symmetry. So the effective action would have the form,

$$\tilde{S} = \int d^d x \left(\frac{1}{2\tilde{g}^2} (\nabla \tilde{\mathbf{n}})^2 + g_1 (\nabla \tilde{\mathbf{n}})^4 + \dots \right) \quad (3.4)$$

Terms like $(\tilde{\mathbf{n}} \cdot \tilde{\mathbf{n}})^k$ also obey $O(N)$ symmetry. However, we are dealing with Goldstone modes, and therefore must have a gapless spectrum. If we start with a weakly coupled theory (small g^2) and integrate out only a tiny fraction of modes (small ζ), then we expect to stay weakly coupled, hence the new theory is also that of a Goldstone mode where only gradient terms are allowed due to Goldstone's theorem.

To carry out the RG procedure, we split the field $\mathbf{n}$ to a low energy and high energy part,

$$\mathbf{n}(x) = \tilde{\mathbf{n}}(x) + \chi(x) \quad (3.5)$$

where,

$$\tilde{\mathbf{n}}(\mathbf{x}) = \int_0^{\frac{\Lambda}{\zeta}} \frac{d^d \mathbf{k}}{(2\pi)^d} e^{i\mathbf{k} \cdot \mathbf{x}} \tilde{\mathbf{n}}(\mathbf{k}) \quad (3.6)$$

and,

$$\chi(\mathbf{x}) = \int_{\frac{\Lambda}{\zeta}}^{\Lambda} \frac{d^d \mathbf{k}}{(2\pi)^d} e^{i\mathbf{k} \cdot \mathbf{x}} \chi(\mathbf{k}) \quad (3.7)$$

Imposing the normalization condition implies

$$\tilde{\mathbf{n}} \cdot \tilde{\mathbf{n}} = 1 \implies \tilde{\mathbf{n}} \cdot \chi = 0 \quad (3.8)$$

Therefore, it is convenient to introduce a basis orthogonal to $\tilde{\mathbf{n}}$ at each $\mathbf{x}$. These are called *frame fields* and we denote them by $\mathbf{e}_\alpha(\mathbf{x})$, where α runs from 1 to $N - 1$. They have the following properties,

$$\begin{aligned} \mathbf{e}_\alpha(\mathbf{x}) \cdot \tilde{\mathbf{n}}(\mathbf{x}) &= 0 \longrightarrow \text{Defining condition} \\ \mathbf{e}_\alpha(\mathbf{x}) \cdot \mathbf{e}_\beta(\mathbf{x}) &= \delta_{\alpha\beta} \longrightarrow \text{Convenient choice} \end{aligned} \quad (3.9)$$

The “dot” is in the N dimensional vector space. Then, we can write the original vector $\mathbf{n}$ as,

$$\mathbf{n}(\mathbf{x}) = \tilde{\mathbf{n}}(\mathbf{x}) \sqrt{1 - \chi^2(\mathbf{x})} + \chi_\alpha(\mathbf{x}) \mathbf{e}_\alpha(\mathbf{x}) \quad (3.10)$$

with,

$$\chi^2 = \chi_\alpha \chi_\alpha \quad (3.11)$$

We will impose one more condition on the basis $\mathbf{e}_\alpha(\mathbf{x})$: that vary in space as smoothly as $\tilde{\mathbf{n}}$. This can always be arranged for because of an $O(N - 1)$ symmetry that we have on the basis. The figure below tries to illustrate this idea. [figure](#). In principle, one could choose a basis at each $\mathbf{x}$ that is arbitrarily oriented. But due to the $O(N - 1)$ symmetry, there is a choice that smoothly extrapolates between all $\mathbf{x}$.

The decomposition is chosen such that both $\mathbf{n}$ and $\tilde{\mathbf{n}}$ are vectors of unit length. The decomposition can be thought of in an alternate way: we choose a basis for the entire N dimensional space, denote by $\hat{\mathbf{e}}^a(\mathbf{x})$, with $a = 1, 2, \dots, N$. We define it to be,

$$\hat{\mathbf{e}}^a(\mathbf{x}) := \{\tilde{\mathbf{n}}(\mathbf{x}), \mathbf{e}_\alpha(\mathbf{x})\} \quad (3.12)$$

In this basis, the original vector can be written as,

$$\mathbf{n}(\mathbf{x}) = u^a(\mathbf{x})\hat{e}^a(\mathbf{x}) \quad (3.13)$$

with,

$$u^a(\mathbf{x}) = \left(\sqrt{1 - \chi^2(\mathbf{x})}, \chi_\alpha(\mathbf{x}) \right) \quad (3.14)$$

Also,

$$\nabla \mathbf{n} = (\nabla u^a)\hat{e}^a + u^a \nabla \hat{e}^a \quad (3.15)$$

expanding $\nabla \hat{e}^a$ using the same basis,

$$\nabla_i \hat{e}^a = \hat{e}^b (\hat{e}^b \cdot \nabla_i \hat{e}^a) \equiv A_i^{ba} \hat{e}^b \quad (3.16)$$

and we get,

$$\nabla_i \mathbf{n} = \hat{e}^a (\nabla_i u^a + A_i^{ab} u^b) \quad (3.17)$$

The introduction of curved coordinates of course changed the definition of the derivative to that of a covariant one. The A field in this context is called a *spin connection* for whatever reasons. It has the property,

$$A_i^{ab} = -A_i^{ba} \quad (3.18)$$

this comes from,

$$\hat{e}^a \cdot \hat{e}^b = \delta^{ab} \quad (3.19)$$

Taking a gradient gives us the result. In this new coordinates, our original action is,

$$S = \int d^d \mathbf{x} \frac{1}{2g^2} (\nabla_i u^a + A_i^{ab} u^b)^2 \quad (3.20)$$

Now let us integrate out the high momentum modes χ . A natural thing that we can do is to expand the above action up-to quadratic terms in χ and do the integral. To leading order, we can then simply replace $\chi^2(\mathbf{x})$ with $\langle \chi^2(\mathbf{x}) \rangle$. ∇u term then will be simply zero. Leading order effective action therefore would be,

$$\tilde{S} = \int d^d \mathbf{x} \frac{1}{2g^2} A_i^{ab} A_i^{ac} \langle u^b u^c \rangle \quad (3.21)$$

From the definition of u^a , we see that the expectation values form a block-diagonal matrix,

$$\langle u^b u^c \rangle = \begin{pmatrix} 1 - \delta_{\alpha\beta} \langle \chi_\alpha \chi_\beta \rangle & 0 \\ 0 & \langle \chi_\alpha \chi_\beta \rangle \end{pmatrix} \quad (3.22)$$

The off-diagonal terms will correspond to odd powers of χ , which averages to zero due to symmetry. We get the action to be,

$$\tilde{S} = \int d^d \mathbf{x} \frac{1}{2g^2} \left[A_i^{\gamma 0} A_i^{\gamma 0} (1 - \delta_{\alpha\beta} \langle \chi_\alpha \chi_\beta \rangle) + (A_i^{0\alpha} A_i^{0\beta} + A_i^{\gamma\alpha} A_i^{\gamma\beta}) \langle \chi_\alpha \chi_\beta \rangle \right] \quad (3.23)$$

Recall that our goal is to find the running of the coupling g . This should appear as the coupling of $\tilde{\mathbf{n}}$ in the new theory. $\tilde{\mathbf{n}}$ enters only through $A_i^{0\alpha}$. The term $A_i^{\alpha\gamma}$ contain only the basis vectors $\mathbf{e}_\alpha$, orthogonal to $\tilde{\mathbf{n}}$. For the purpose of finding the running of g , we can ignore them now. We have,

$$A_i^{\gamma 0} A_i^{\gamma 0} = (\mathbf{e}_\gamma \cdot \nabla_i \tilde{\mathbf{n}})(\mathbf{e}_\gamma \cdot \nabla_i \tilde{\mathbf{n}}) = \nabla_i \tilde{\mathbf{n}} \cdot \nabla_i \tilde{\mathbf{n}} \quad (3.24)$$

We can also deduce the following: due to the $O(N-1)$ symmetry, we must have,

$$\langle \chi_\alpha \chi_\beta \rangle = \delta_{\alpha\beta} I_d \quad (3.25)$$

where I_d is a dimension d dependent quantity that we will compute soon. So for $\tilde{\mathbf{n}}$, the effective action is,

$$\tilde{S}_{\frac{\Lambda}{\zeta}} = \int d^d \mathbf{x} \frac{1}{2g^2} [1 - (N-1)I_d + I_d] (\nabla \tilde{\mathbf{n}})^2 \quad (3.26)$$

We can now perform the second step of RG, which is re-scaling $\mathbf{x} \rightarrow \mathbf{x}' = \frac{\mathbf{x}}{\zeta}$ so that the effective field theory for $\tilde{\mathbf{n}}$ also has the same cut-off Λ . Then,

$$\tilde{S}_\Lambda = \int d^d \mathbf{x}' \zeta^{d-2} \left(\frac{1}{2g^2} + \frac{2-N}{2g^2} I_d \right) (\nabla' \tilde{\mathbf{n}})^2 \quad (3.27)$$

giving us,

$$\frac{1}{g^2(\zeta)} = \zeta^{d-2} \left(\frac{1}{g^2} + \frac{2-N}{g^2} I_d(\zeta) \right) \quad (3.28)$$

All that is left is to find I_d , which is just the propagator for χ . To find this, we need the free action for the χ field. This has to come from the square of $\nabla_i u^a$ in the action, as the $A \cdot u$ term is describing how χ couples to the other fields $\tilde{\mathbf{n}}$ and the frame fields. This is extremely simple, the free action for χ is then,

$$S_{\text{free}}[\chi] = \int d^d \mathbf{x} \frac{1}{2g^2} (\nabla \chi_\alpha)^2 \quad (3.29)$$

the propagator is simply the inverse, and we need it evaluated at equal position,

$$\begin{aligned} \langle \chi_\alpha(\mathbf{x}) \chi_\beta(\mathbf{x}) \rangle &= g^2 \delta_{\alpha\beta} \int_{\frac{\Lambda}{\zeta}}^{\Lambda} \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{\mathbf{k}^2} \\ &= g^2 \Lambda^{d-2} \delta_{\alpha\beta} \frac{\Omega_{d-1}}{(2\pi)^d} \begin{cases} \zeta - 1 & \text{if } d = 1 \\ \ln(\zeta) & \text{if } d = 2 \\ 1 - \zeta^{d-2} & \text{if } d > 2 \end{cases} \end{aligned} \quad (3.30)$$

For ζ close to unity, and in the weakly coupled limit ($g^2 \Lambda^{d-2}$ small), we can consistently do what we did so far: that is expand to quadratic order in χ and discard higher order terms.

For $d = 1$, g^2 is relevant and increase as we go to IR. As g^2 is proportional to the inverse of the squared average magnetization, we see that the average magnetization go to zero in the IR, that is we end up at a disordered phase. The question is: what happens to the Goldstone mode? Since the theory become strongly coupled, we cannot in any way compute the partition function and see what happens to the Goldstone modes. It turns out that in IR, the Goldstone modes get gapped. This will be shown in an example problem.

For $d > 2$, g^2 decrease as we go to IR, and therefore the massless Goldstone modes described by the non-linear sigma model is a good description of the long distance physics, and we indeed have symmetry breaking and therefore long range order.

$d = 2$ is the most interesting case. Here,

$$\frac{1}{g^2(\zeta)} = \frac{1}{g^2} + \frac{2-N}{2\pi} \ln(\zeta) \quad (3.31)$$

It is customary to re-write this using $\zeta = e^s$. What we have done is so far is RG with small s . Therefore, it makes sense to look at the RG flow as a differential equation. Taking a derivative with respect to s on both sides,

$$\frac{dg(s)}{ds} \equiv \beta(g) = \frac{N-2}{4\pi} g^3(s) \quad (3.32)$$

For $N = 1$, g^2 decreases as we go to IR, meaning Goldstone modes are a good description and there is an ordered phase. This confirms our Ising model result in two dimensions, at least qualitatively. For $N > 2$, the theory becomes strongly coupled, which imply disorder – a lack of phase transitions. Such theories which are strongly coupled in the IR (therefore weakly coupled in UV) are called *asymptotically free*. For $N > 3$ the theory of

the Goldstone modes is no longer solvable, and what becomes of the Goldstone modes is something that we will answer below. $N = 2$ seems rather special, and this will need extra care. It turns out that for $N = 2$, there is a different kind of phase transition – due to topology – that we will discuss later.

3.2. Large N

Let us now answer the question of the IR physics of the $d = 2$ Goldstone modes. As we saw, in IR for $N > 2$, the Goldstone modes become strongly coupled, and we can no longer apply perturbation theory. But a physicist only has perturbation theory at their disposal. So we would be desperate and invent a perturbative parameter. We will take N to be really large, so $\frac{1}{N}$ becomes our artificial dimensionless small parameter. We re-write the partition function as,

$$\begin{aligned} Z &= \int \mathcal{D}\mathbf{n} \, \delta(\mathbf{n}^2 - 1) \exp\left(-\frac{1}{2g^2} \int d^d\mathbf{x} \, (\nabla\mathbf{n})^2\right) \\ &= \int \mathcal{D}\mathbf{n} \int \mathcal{D}\sigma \exp\left(-\frac{1}{2g^2} \int d^d\mathbf{x} \, (\nabla\mathbf{n})^2 - \frac{i}{2g^2} \int d^d\mathbf{x} \, \sigma(\mathbf{n}^2 - 1)\right) \end{aligned} \quad (3.33)$$

In the second step, we just re-wrote the delta function as a functional integral over the auxiliary field σ . The $\frac{1}{2g^2}$ for the sigma part as well is achieved by a simple re-scaling of σ , and this is done for convenience. We can now actually integrate out $\mathbf{n}$, to get,

$$\begin{aligned} Z &= \int \mathcal{D}\sigma \, [\det(-\nabla^2 + i\sigma)]^{-\frac{N}{2}} \exp\left(\frac{i}{2g^2} \int d^d\mathbf{x} \, \sigma(\mathbf{x})\right) \\ &= \int \mathcal{D}\sigma \exp\left(-\frac{N}{2} \text{tr} \ln(-\nabla^2 + i\sigma) + \frac{i}{2g^2} \int d^d\mathbf{x} \, \sigma(\mathbf{x})\right) \end{aligned} \quad (3.34)$$

All of these steps do not care whether the theory is strongly coupled or not. For g^2 large, we are still unable to do the σ integral. However, the N in front of the first term gives us an idea: we could evaluate the integral as a saddle point if we take $N \rightarrow \infty$. This seems almost possible, except the second term do not have an N . We will invent an N as follows,

$$\tilde{g}^2 = Ng^2 \quad (3.35)$$

such that we keep $\tilde{g}$ finite as we send $N \rightarrow \infty$. This naively feels like against the spirit of the idea. To keep $\tilde{g}$ finite, one must send $g \rightarrow 0$, which is weakly coupled theory. But, the running of $\tilde{g}$ is,

$$\frac{1}{\tilde{g}^2(\zeta)} = \frac{1}{\tilde{g}^2} + \frac{1}{2\pi} \ln(\zeta) \quad \text{for large } N \quad (3.36)$$

meaning the coupling $\tilde{g}$ grows as we go to IR. We will nonetheless demand that $\tilde{g}$ is finite for $N \rightarrow \infty$, albeit large. So we will still end up with a strongly coupled theory in IR for large N , just that the coupling is now $\tilde{g}$.

$$Z = \int \mathcal{D}\sigma \exp \left[-\frac{N}{2} \left(\text{tr} \ln(-\nabla^2 + i\sigma) + \frac{i}{2\tilde{g}^2} \int d^d \mathbf{x} \sigma(\mathbf{x}) \right) \right] \quad (3.37)$$

can be evaluated as a saddle point for $N \rightarrow \infty$. We find the equation of motion for σ ,

$$\text{tr} \left((-\nabla^2 + i\sigma)^{-1} i\delta\sigma \right) = \int d^d \mathbf{x} \frac{i}{\tilde{g}^2} \delta\sigma(\mathbf{x}) \quad (3.38)$$

the trace will be evaluated in position space,

$$\int d^d \mathbf{x} \langle \mathbf{x} | \frac{1}{-\nabla^2 + i\sigma(\mathbf{x})} | \mathbf{x} \rangle i\delta\sigma(\mathbf{x}) = \int d^d \mathbf{x} \frac{i}{\tilde{g}^2} \delta\sigma(\mathbf{x}) \quad (3.39)$$

we have the equation of motion,

$$\langle \mathbf{x} | \frac{1}{-\nabla^2 + i\sigma(\mathbf{x})} | \mathbf{x} \rangle = \frac{1}{\tilde{g}^2} \quad (3.40)$$

The left hand side must be real and independent of $\mathbf{x}$ for the above to make sense. So, we choose,

$$\sigma(\mathbf{x}) = -i\mu^2 \quad (3.41)$$

and get,

$$\langle \mathbf{x} | \frac{1}{-\nabla^2 + \mu^2} | \mathbf{x} \rangle = \int_0^\Lambda \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{\mathbf{k}^2 + \mu^2} = \frac{1}{\tilde{g}^2} \quad (3.42)$$

For $d = 2$,

$$\frac{1}{\tilde{g}^2} = \frac{1}{2\pi} \int_0^\Lambda dk \frac{k}{k^2 + \mu^2} = \frac{1}{4\pi} \ln \left(1 + \frac{\Lambda^2}{\mu^2} \right) \quad (3.43)$$

giving us a solution for μ to be,

$$\mu^2 \approx \Lambda^2 \exp \left(-\frac{4\pi}{\tilde{g}^2} \right) \quad (3.44)$$

Here, we have assumed $\mu \ll \Lambda$, and this is consistent when $\tilde{g}^2 \ll 1$. For very weakly coupled theory, we see that μ is close to zero. However, when $\tilde{g}^2$ become large, μ^2 becomes large

(still exponentially smaller than Λ , but not close to zero). From Equation (3.33), we see that non-zero value for σ field imply a gap for $\boldsymbol{n}$. The gap opens as we go deeper into IR. This is the fate of the Goldstone modes in $d = 2$ at least for large N . They become gapped. The chain of ideas is as follows:

$$\begin{aligned}
d = 2, N \gg 1 &\rightarrow \text{strong coupling in IR} \\
&\rightarrow \text{no ordered phase} \\
&\rightarrow \text{gap for Goldstone modes}
\end{aligned} \tag{3.45}$$

From numerics, it turns out that the same conclusions hold for $N \geq 3$.

3.3. References and Further Reading

- The non-linear sigma model analyzed in Section 3.1 was done by Polyakov in a whooping two pages in [11]. It is also explained in section 13.3 of [7], but that is a bit confusing and cumbersome.

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